Forecast Accuracy Measurement. Store Clusterization

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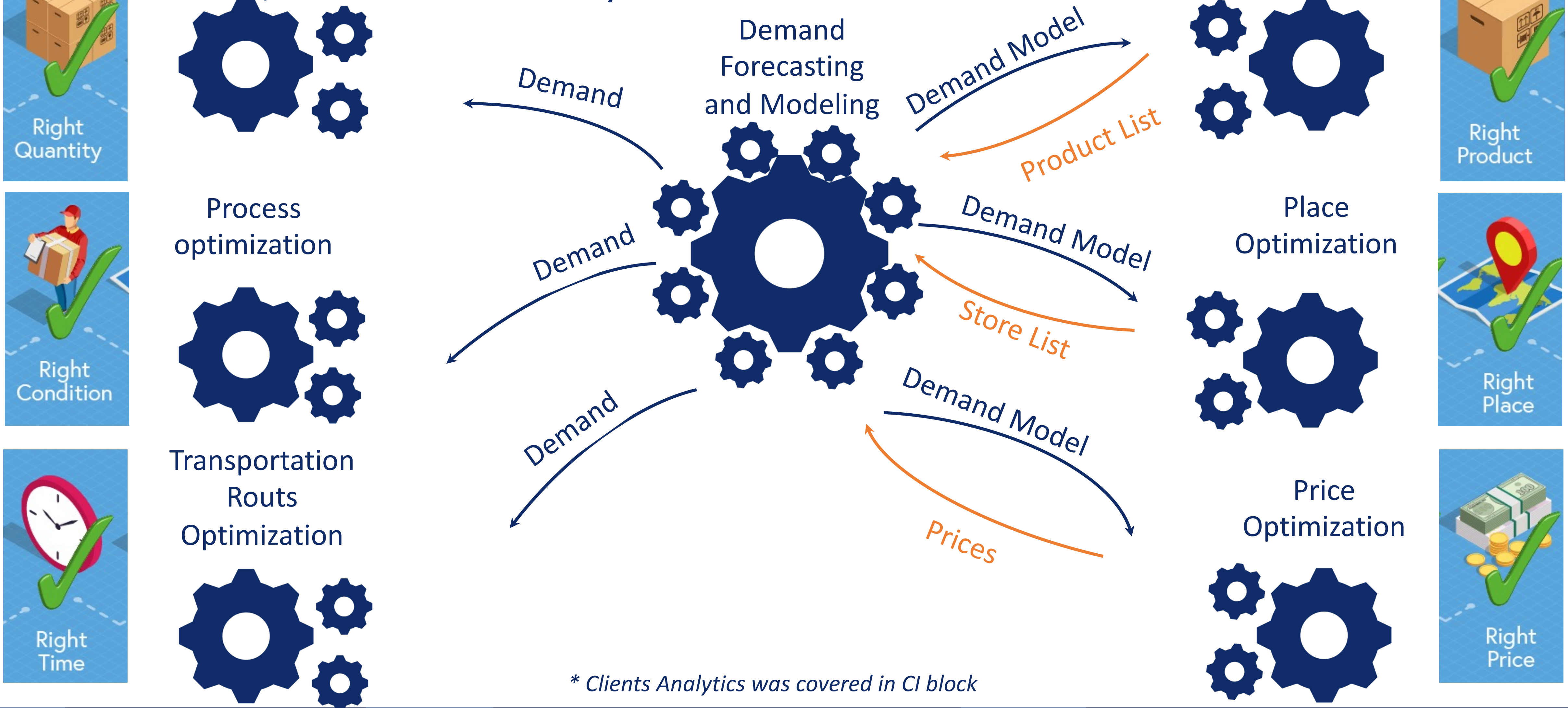
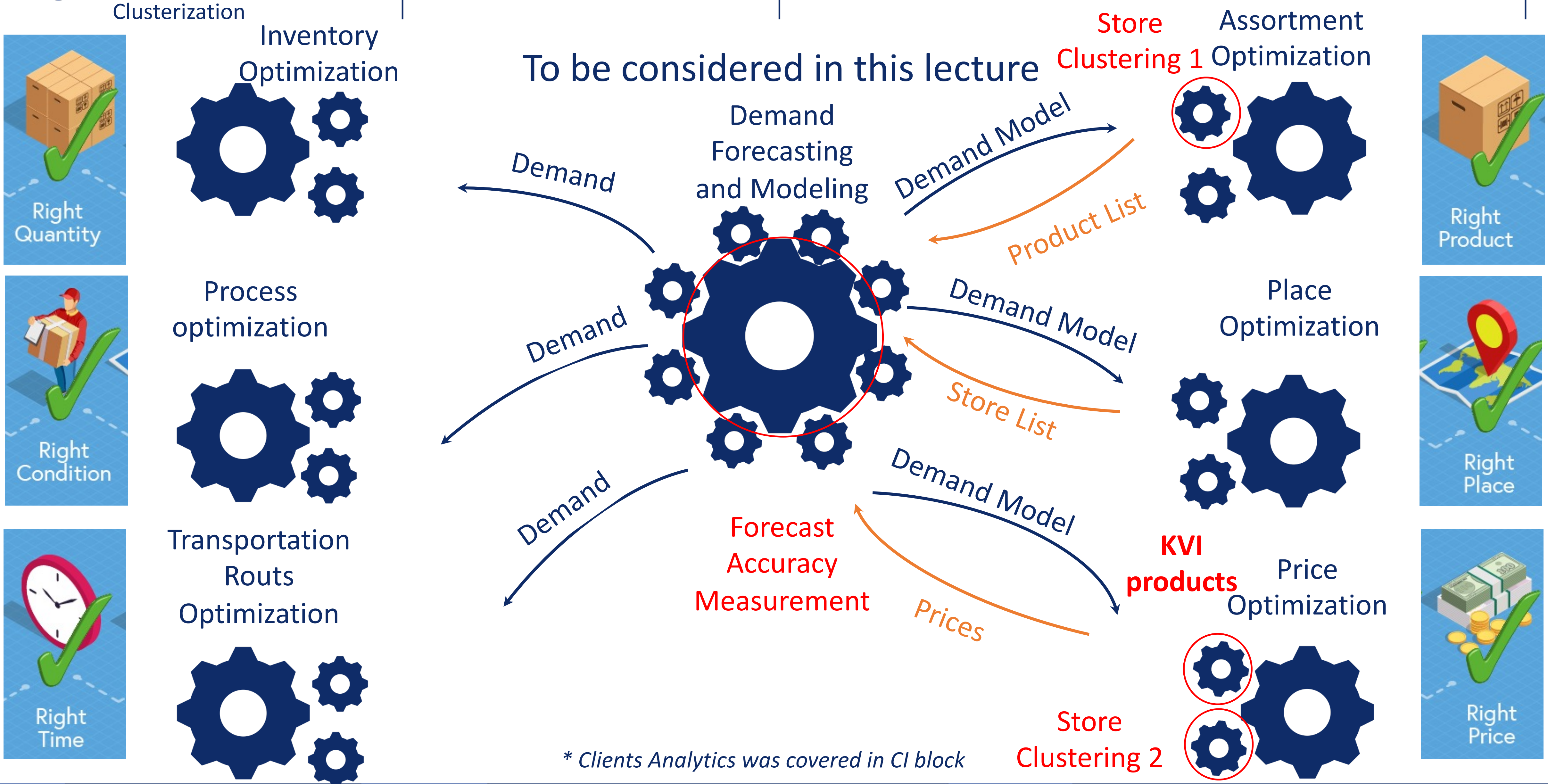
Agenda

- Forecast Accuracy Measurement
- Store Clusterization in Retail
- Building KVI products
- HW #3
    - Demand Forecasting

Analytical tasks in Retail Chain*



** Clients Analytics was covered in CI block*


To be considered in this lecture
Inventory Optimization
Right Quantity
Process optimization
Right Condition
Transportation Routs Optimization
Right Time
Demand
Demand
Demand
Demand Forecasting and Modeling
Forecast Accuracy Measurement
Demand Model
Product List
Store Clustering 1
Assortment Optimization
Right Product
Demand Model
Store List
Place Optimization
Right Place
Demand Model
Prices
KVI products
Price Optimization
Store Clustering 2
Right Price
* Clients Analytics was covered in CI block

How to Best Measure.. ..Forecast Error

Accuracy Business Measures

	id1	id2	id3	id4
Actual	1	10	10	1
Forecast	10	9	1	2

Items					
id1					
Id2					
Id3					
Id4					
all					

Accuracy Business Measures

	id1	id2	id3	id4
Actual	1	10	10	1
Forecast	10	9	1	2

Items	Mean Squared Error MSE $1/\mathbf{n} \cdot \sum(F-A)^2$				
id1	81				
Id2	1				
Id3	81				
Id4	1				
all	**41**				

Accuracy Business Measures

	id1	id2	id3	id4
Actual	1	10	10	1
Forecast	10	9	1	2

Items	Mean Squared Error MSE $1/\mathbf{n} \cdot \sum(F - A)^2$	Mean Absolute Error MAE $1/\mathbf{n} \cdot \sum\lvert F - A\rvert$			
id1	81	9			
Id2	1	1			
Id3	81	9			
Id4	1	1			
all	**41**	**5**			

Accuracy Business Measures

	id1	id2	id3	id4
Actual	1	10	10	1
Forecast	10	9	1	2

Items	Mean Squared Error MSE $1/\mathbf{n} \cdot \sum(F - A)^2$	Mean Absolute Error MAE $1/\mathbf{n} \cdot \sum\lvert F - A\rvert$	MA Percentage Error MAPE $1/\mathbf{n} \cdot \sum(\lvert F - A\rvert/A)$		
id1	81	9	900%		
Id2	1	1	10%		
Id3	81	9	90%		
Id4	1	1	100%		
all	**41**	**5**	**275%**		

Accuracy Business Measures

	id1	id2	id3	id4
Actual	1	10	10	1
Forecast	10	9	1	2

Items	Mean Squared Error MSE $1/\mathbf{n} \cdot \sum(F-A)^2$	Mean Absolute Error MAE $1/\mathbf{n} \cdot \sum\lvert F-A\rvert$	MA Percentage Error MAPE $1/\mathbf{n} \cdot \sum(\lvert F-A\rvert/A)$	Weighted APE WAPE $\sum\lvert F-A\rvert/\sum\lvert A\rvert$	
id1	81	9	900%	900%	
Id2	1	1	10%	10%	
Id3	81	9	90%	90%	
Id4	1	1	100%	100%	
all	**41**	**5**	**275%**	**20/22*100%**	

MAPE vs WAPE

$$MAPE = \frac{1}{n}\sum_{i:\, SKU,TT}\frac{|F_i - A_i|}{A_i} = \sum_{SKU,TT}\frac{|F_i - A_i|}{A_i}\cdot\frac{1}{n}$$

MAPE vs WAPE

$$MAPE = \frac{1}{n}\sum_{i:\ SKU,TT} \frac{|F_i - A_i|}{A_i} = \sum_{SKU,TT} \frac{|F_i - A_i|}{A_i} \cdot \frac{1}{n}$$

$$w_i$$

MAPE vs WAPE

$$MAPE = \frac{1}{n}\sum_{i:\,SKU,TT}\frac{|F_i - A_i|}{A_i} = \sum_{SKU,TT}\frac{|F_i - A_i|}{A_i}\cdot\frac{1}{n}$$

w_i

This is a weight of an item i

MAPE vs WAPE

$$MAPE = \frac{1}{n}\sum_{i:\,SKU,TT} \frac{|F_i - A_i|}{A_i} = \sum_{SKU,TT} \frac{|F_i - A_i|}{A_i} \cdot \frac{1}{n}$$

w_i

This is a weight of an item i

In WAPE case a weight of an item is proportional to its actual volume:

$$w_i = \frac{A_i}{\sum_{j:SKU,TT} A_j}$$

MAPE vs WAPE

$$MAPE = \frac{1}{n}\sum_{i:\,SKU,TT}\frac{|F_i - A_i|}{A_i} = \sum_{SKU,TT}\frac{|F_i - A_i|}{A_i}\cdot\frac{1}{n}$$

w_i

This is a weight of an item i

In WAPE case a weight of an item is proportional to its actual volume:

$$w_i = \frac{A_i}{\sum_{j:SKU,TT} A_j}$$

$$WAPE = \sum_{i:SKU,TT}\frac{|F_i - A_i|}{A_i}\cdot w_i = \sum_{i:SKU,TT}\frac{|F_i - A_i|}{A_i}\cdot\frac{A_i}{\sum_{j:SKU,TT} A_j} = \frac{\sum_{i:SKU,TT}|F_i - A_i|}{\sum_{j:SKU,TT} A_j}$$

Accuracy Business Measures

	id1	id2	id3	id4
Actual	1	10	10	1
Forecast	10	9	1	2

Items	Mean Squared Error MSE $1/\mathbf{n} \cdot \sum(F-A)^2$	Mean Absolute Error MAE $1/\mathbf{n} \cdot \sum\|F-A\|$	MA Percentage Error MAPE $1/\mathbf{n} \cdot \sum(\|F-A\|/A)$	Weighted APE WAPE $\sum\|F-A\|/\sum\|A\|$	WA Predicted PE WAPPE $\sum\|F-A\|/\sum\|F\|$
id1	81	9	900%	900%	90%
Id2	1	1	10%	10%	1/9*100%
Id3	81	9	90%	90%	900%
Id4	1	1	100%	100%	50%
all	**41**	**5**	**275%**	**20/22*100%**	**20/22*100%**

Accuracy Business Measures

$$MAE = \sum |F - A|$$

+ Consider weight of items in the company
+ Doesn't stimulate nor Overforecasting or UnderForecasting Issues
– Lack of interpretability (it is nor clear whether is good or not when MAE = 10000)

$$MAPE = \frac{1}{n} \sum \frac{|F-A|}{|A|}$$

+ Is interpretable
+ Marks cases with abnormally high losses (can be more than 100%)
– Doesn't Consider weight of items in the company
– Stimulates UnderForecasting Issues

$$WAPE = \frac{\sum |F-A|}{\sum |A|}$$

+ Is interpretable
+ Consider weight of items in the company
+ Marks cases with abnormally high losses (can be more than 100%)
– Stimulates UnderForecasting Issues

Accuracy Business Measures

$$\text{WAPPE}=\frac{\sum(|F-A|)}{\sum F}$$

+ Is interpretable
+ Consider weight of items in the company
+ Marks cases with abnormally high losses (can be more than 100%)
+ Stimulates OverForecasting Issues

$$\text{SWAPE}=2\frac{\sum(|F-A|)}{\sum F+\sum A}$$

+ Is interpretable
+ Consider weight of items in the company
+ Doesn't stimulate nor Overforecasting or UnderForecasting Issues
+ Accuracy = 1-WAMAXPE
– Doesn't mark cases with abnormally high losses (can not be more than 100%)

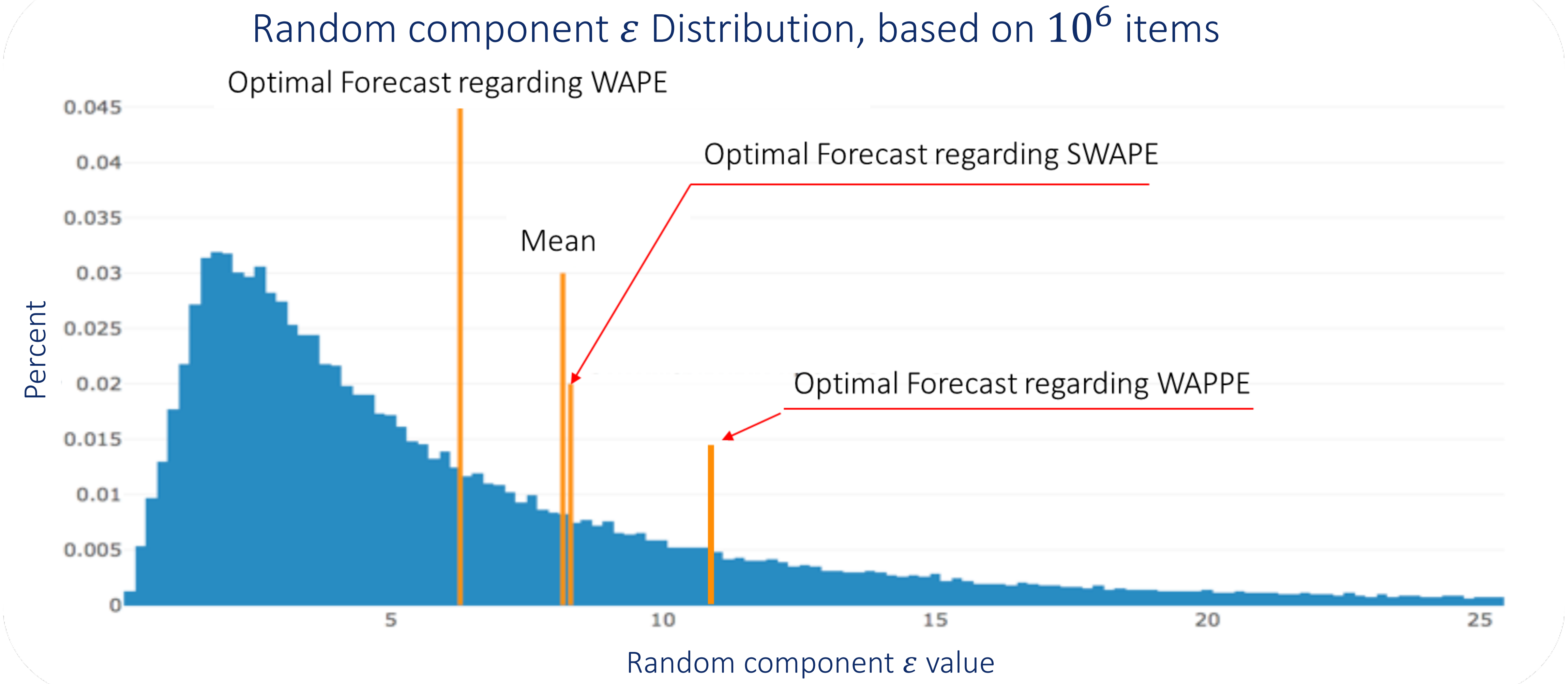
$$\text{WAMAXPE}=\frac{\sum(|F-A|)}{\sum \max(|A|,|F|)}$$

+ Is interpretable
+ Consider weight of items in the company
+ Doesn't stimulate nor Overforecasting or UnderForecasting Issues
+ Accuracy = 1-WAMAXPE
+ Doesn't mark cases with abnormally high losses (can not be more than 100%)

Selection of the Forecast Accuracy Criterion

- Questions:
  - Does Accuracy Criterion affect on Forecast Values?
    - ✓ Yes
  - Does Accuracy Criterion affect on the Best Forecasting Approach (Algorithm)?
    - ✓ Yes
  - If Top managers decided to change Forecast Accuracy Criterion, will it require the re-estimation of forecasting algorithms?
    - ✓ Yes

Forecast Value Highly Depends on Accuracy Measure. Experiment on real data

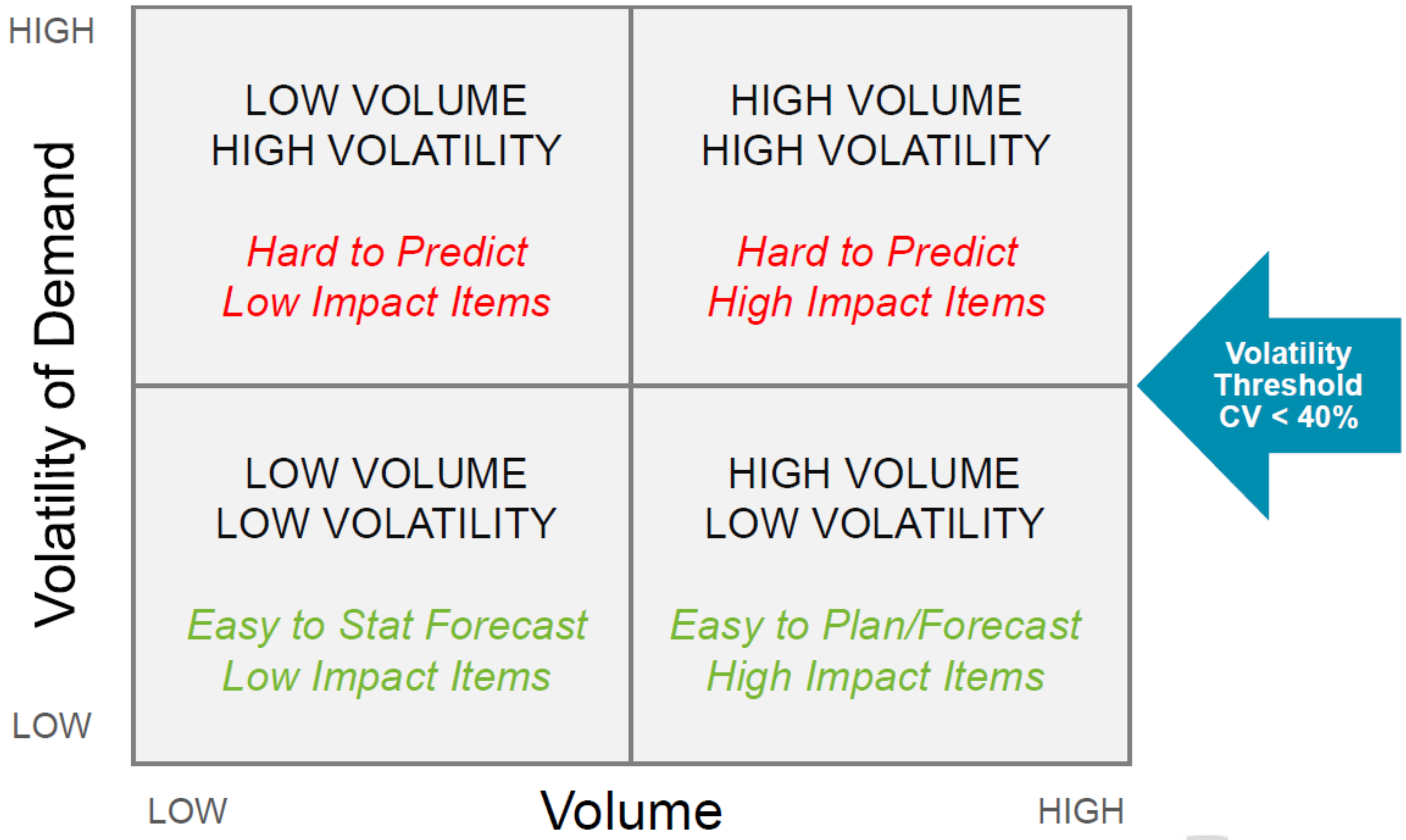


Demand = Deterministic Component + $Random$

$$\text{Demand} = \text{Deterministic Component} + Random$$

$$\mathrm{CV}_i = \mathrm{Var}(Random_i)/\text{Deterministic\_Component}$$

Product/Location Segmentation approach



How to define CV Threshold? EM-based clustering

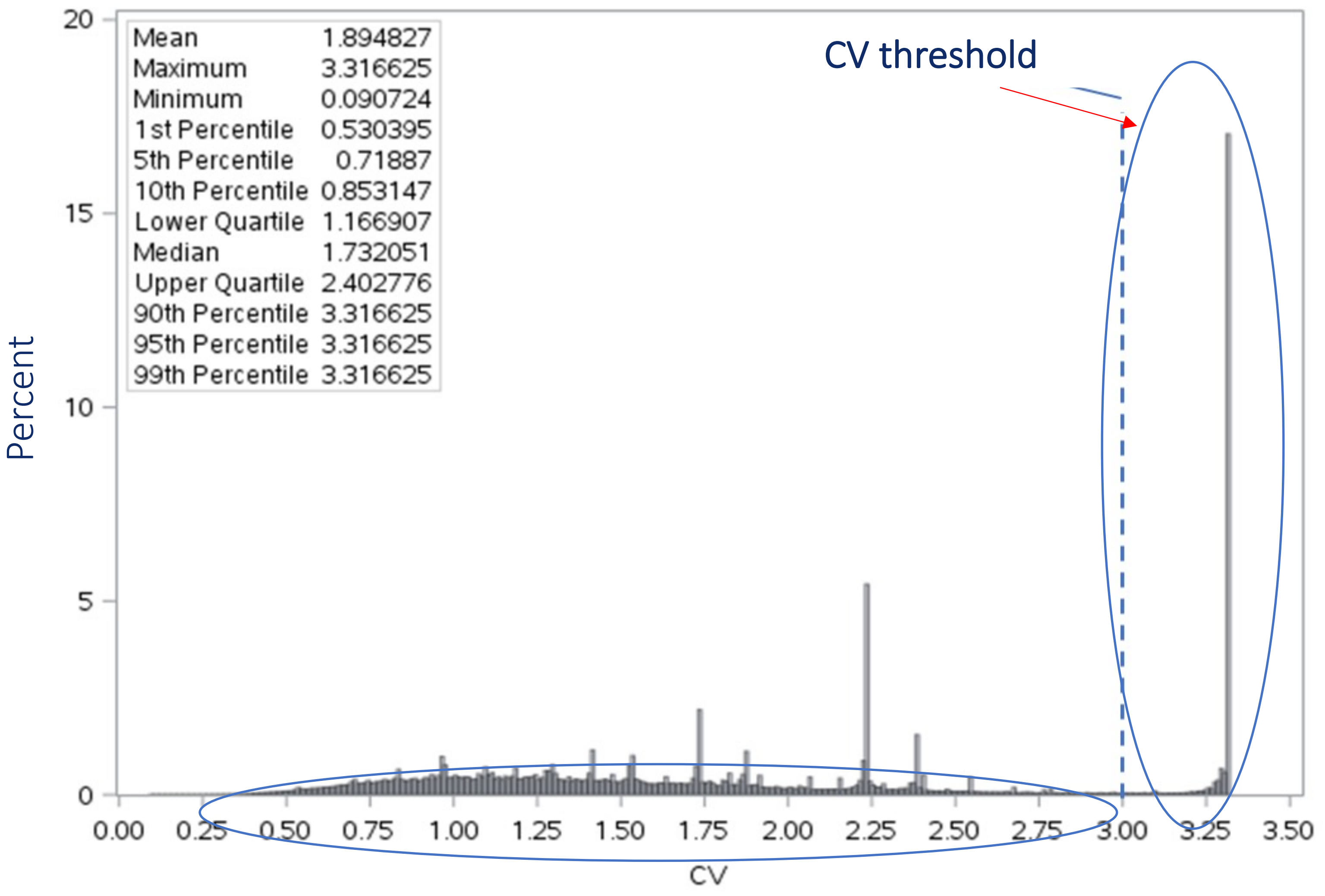
- Approach #1:

  Build bar chart and define threshold manually

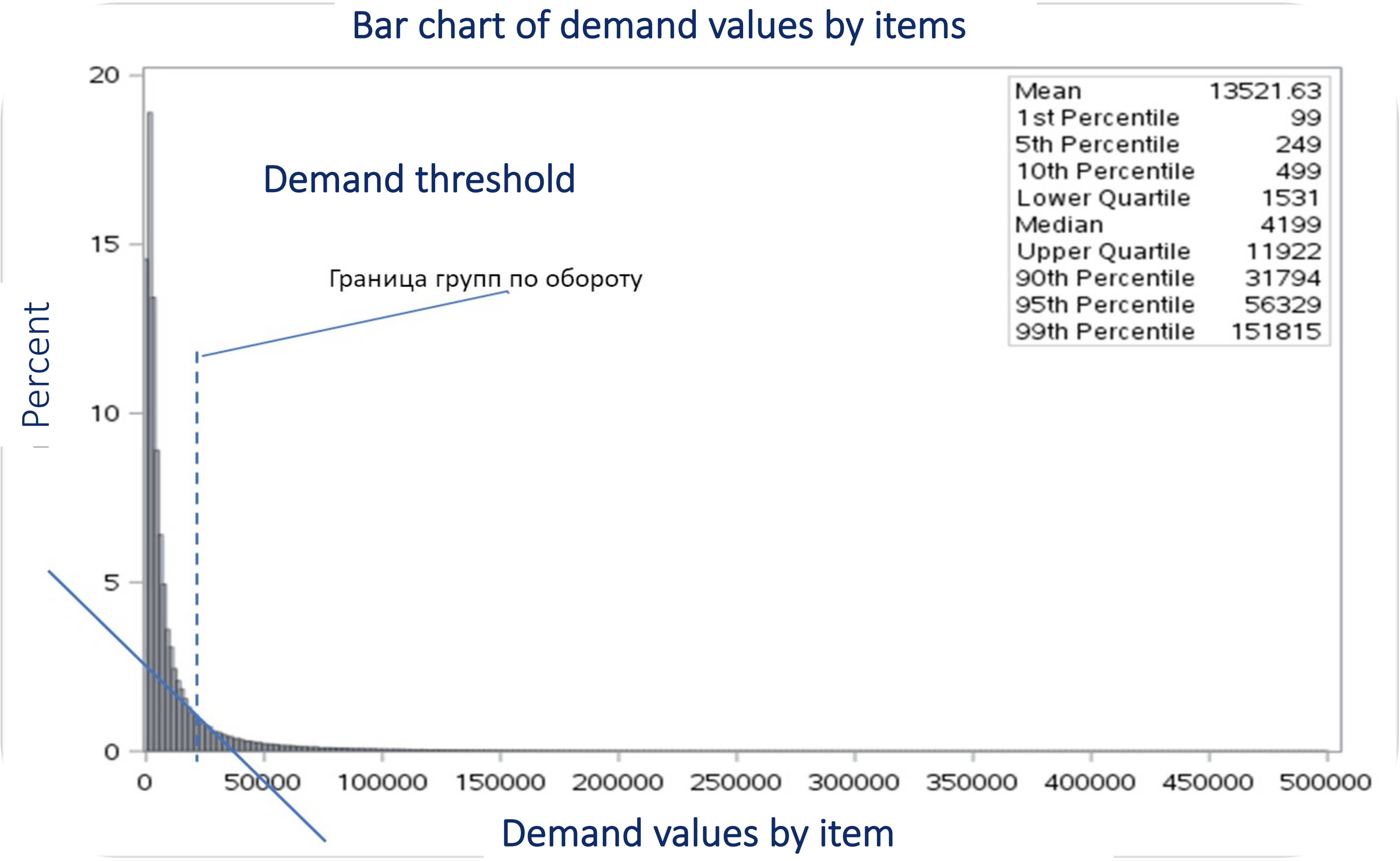
- Approach #2

  Run EM algorithm on CV values (1-column train set)

Bar chart of CV values of different items



How to define Demand Threshold



- Method of tangents
- Method of polygon
- Method of loop
- Method of triangle (the most flexible one)

How to set up Benchmark for Accuracy?

WAPE = >=50%

WAPE = 40%-50%

Volatility of Demand

LOW VOLUME HIGH VOLATILITY *Hard to Predict* *Low Impact Items*	HIGH VOLUME HIGH VOLATILITY *Hard to Predict* *High Impact Items*
LOW VOLUME LOW VOLATILITY *Easy to Stat Forecast* *Low Impact Items*	HIGH VOLUME LOW VOLATILITY *Easy to Plan/Forecast* *High Impact Items*

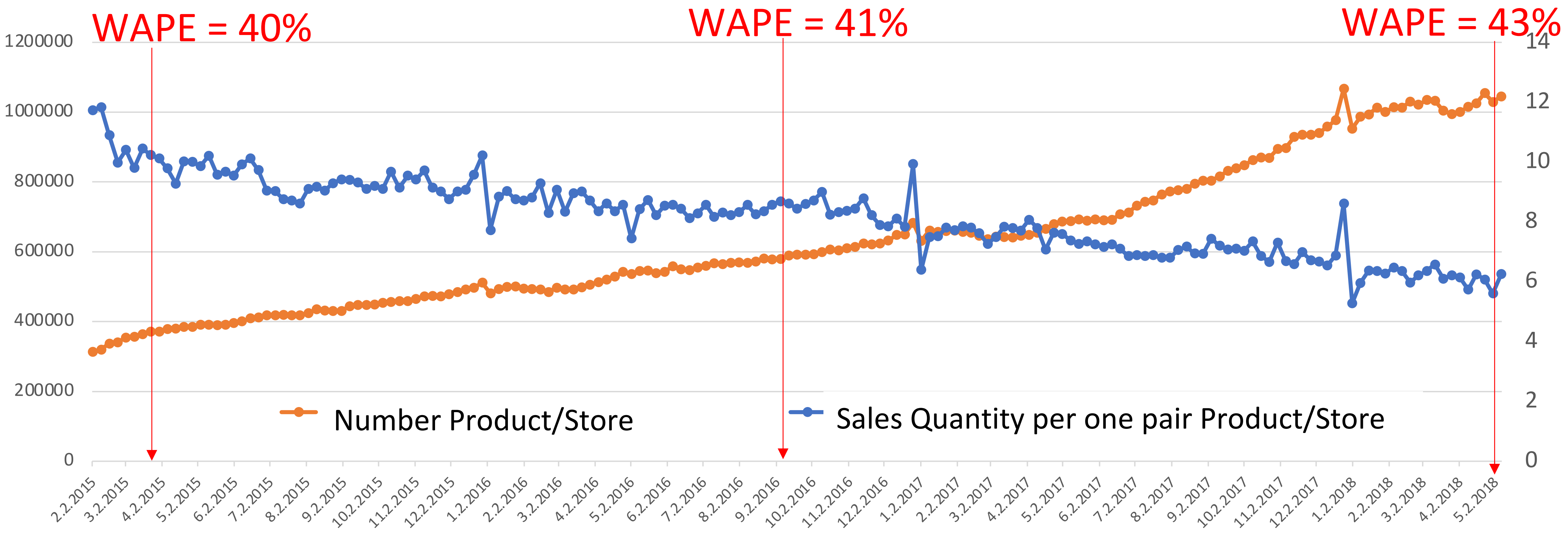
LOW — Volume — HIGH

Volatility Threshold CV < 40%

WAPE = 20%-30%

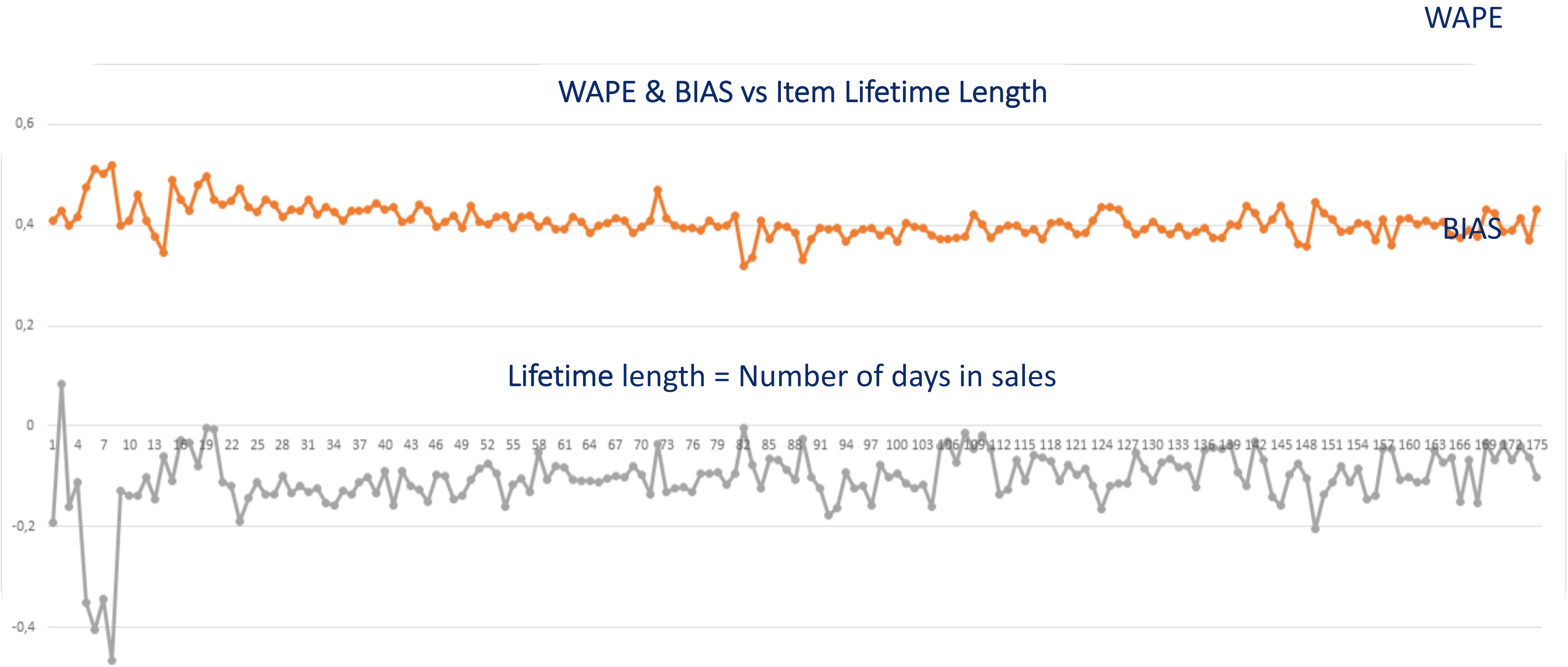
WAPE = 10%-20%

Accuracy dependance on Assortment Dynamics

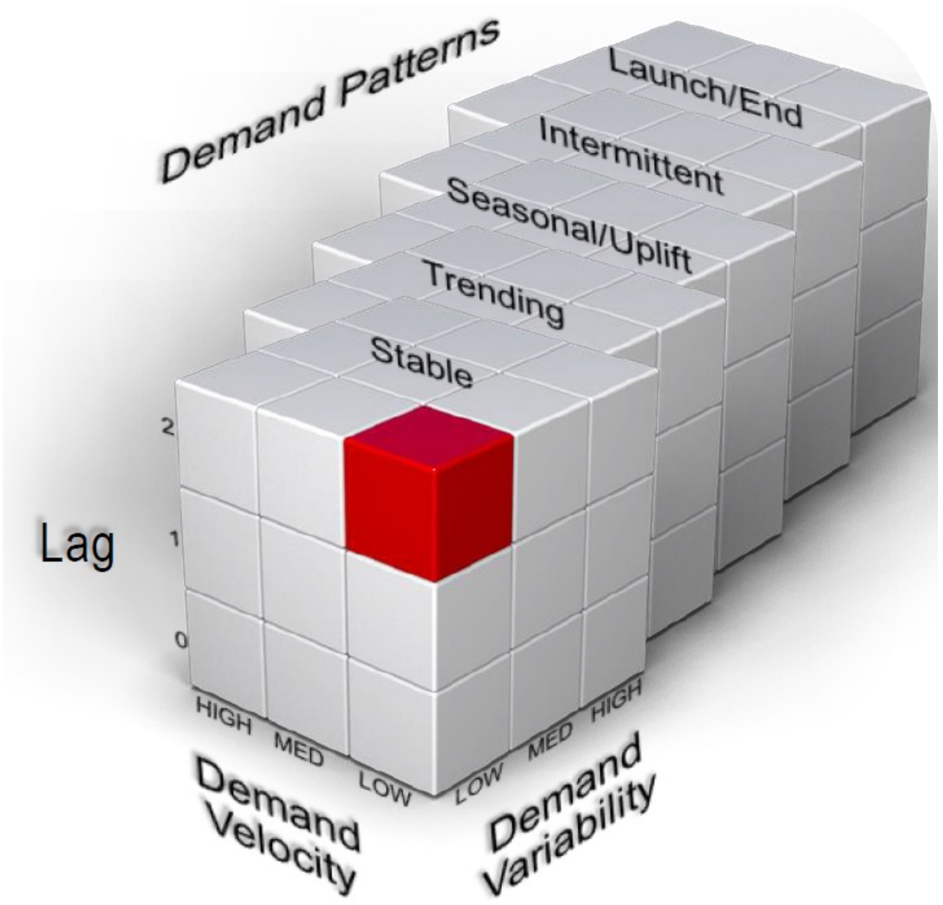


Dynamics of Number of active pairs Product / Store in the Assortment Matrix

Accuracy Depending on of the Sales History Length



Demand Forecasting Units Segmentation based on Forecastability



Agenda

- Forecast Accuracy Measurement
- Store Clusterization in Retail
- Building KVI products
- HW #3
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Определение корзин KVI

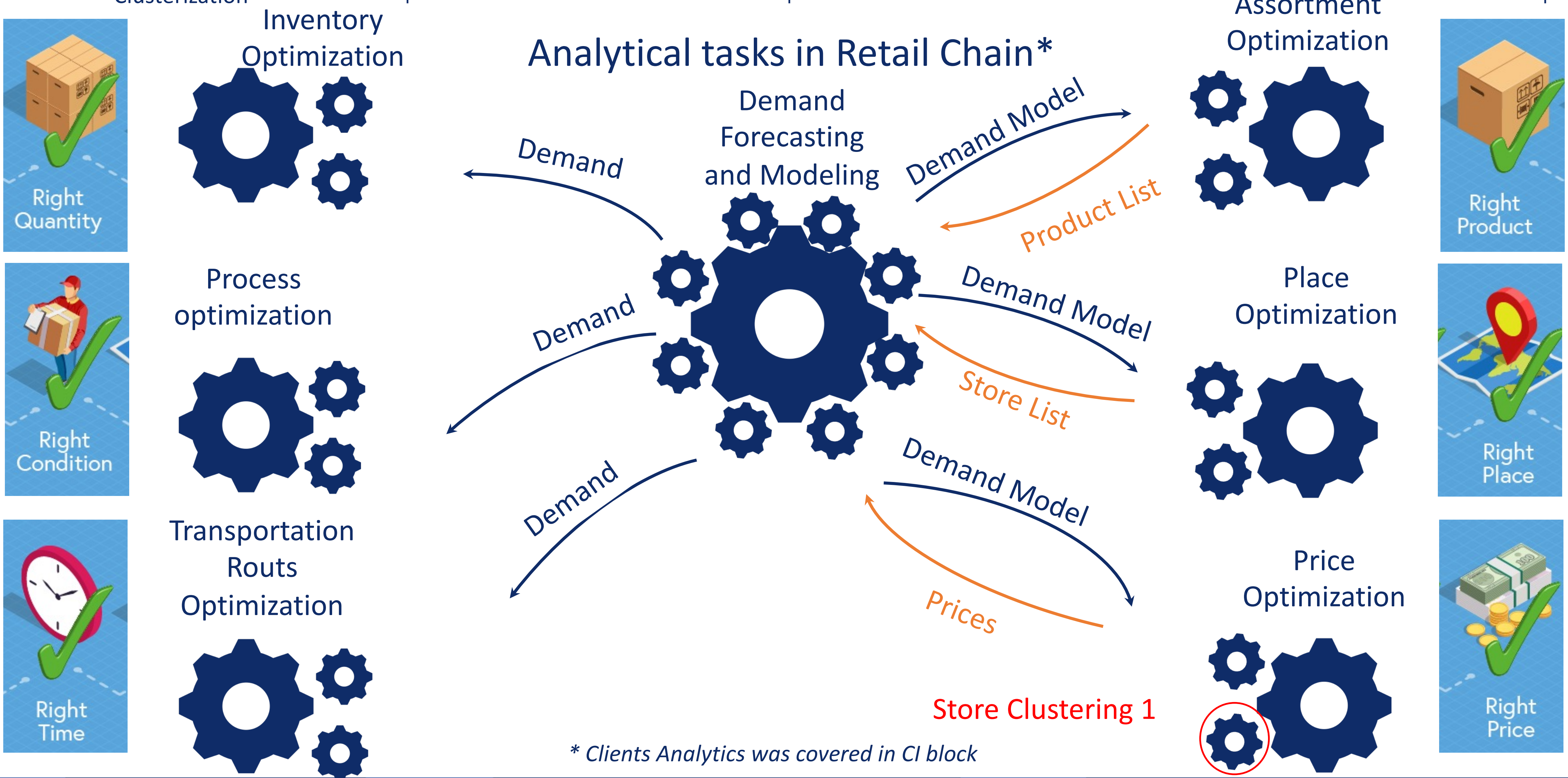
Постановка задачи:

1. Определить список KVI товаров, вычислив KVI Score критерии
2. Определить оптимальный порог для KVI Score критерия

Входные данные:

1. Мониторинг цен конкурентов
2. Чековые данные с 1 января по 31 марта 2018 года
3. Список ТПЦ и СТМ

Analytical tasks in Retail Chain*

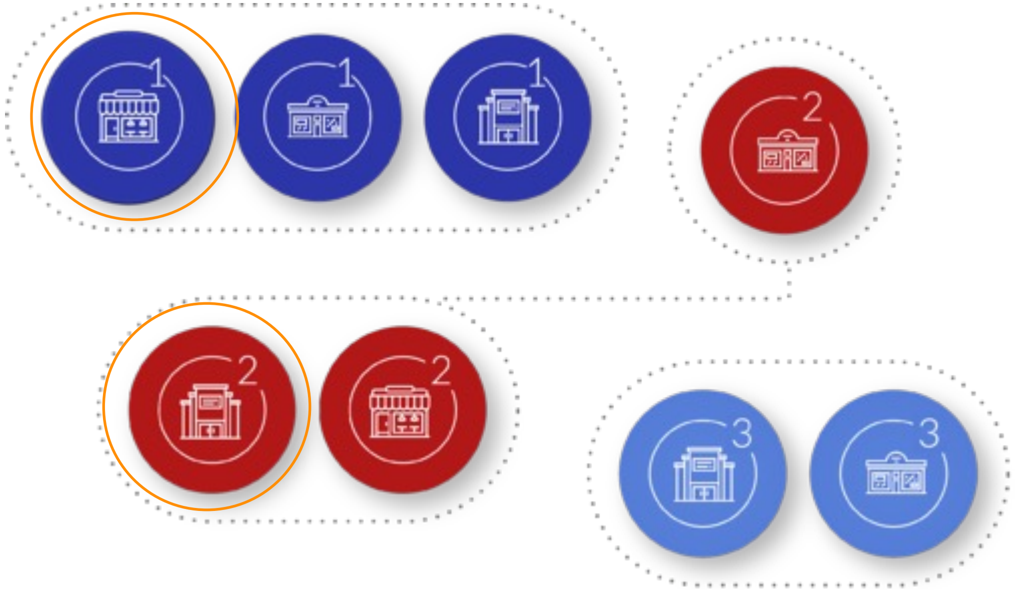


** Clients Analytics was covered in CI block*

Requirement (Ideas) of Store Clusterization for Price Optimization

Why in different price-clusters?

- different Competitor Environment
- different Demand Structure
- different Clients



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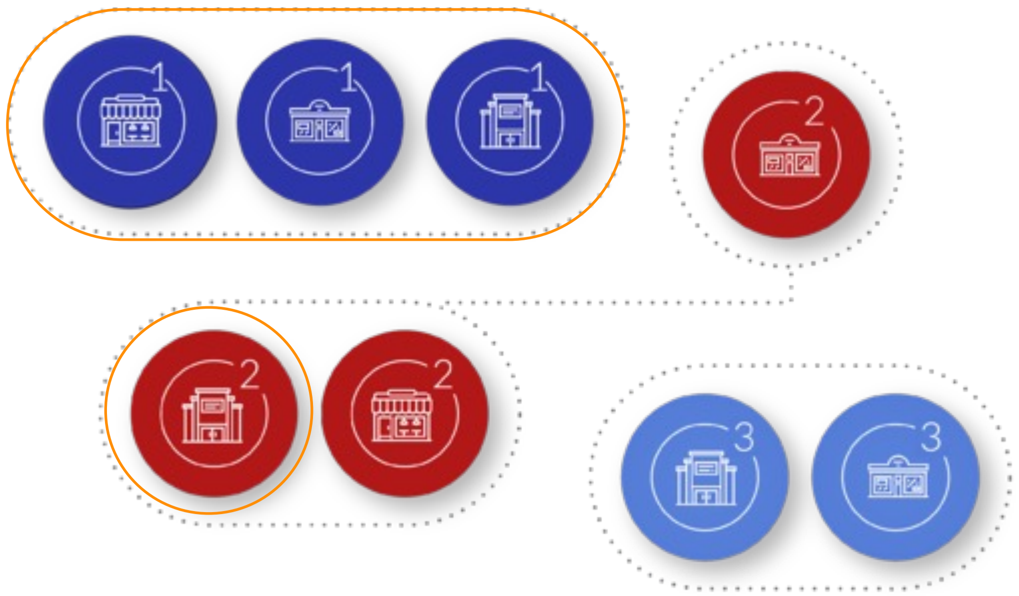
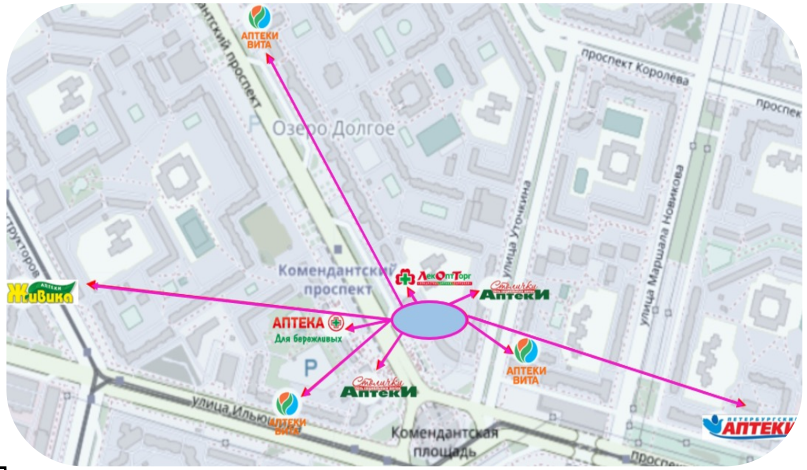


Why in the same price-cluster?

- the same Competitors and/or Demand Structure and/or Clients
- located close to each other (geolocation)

Clustering based on Competitive Environment

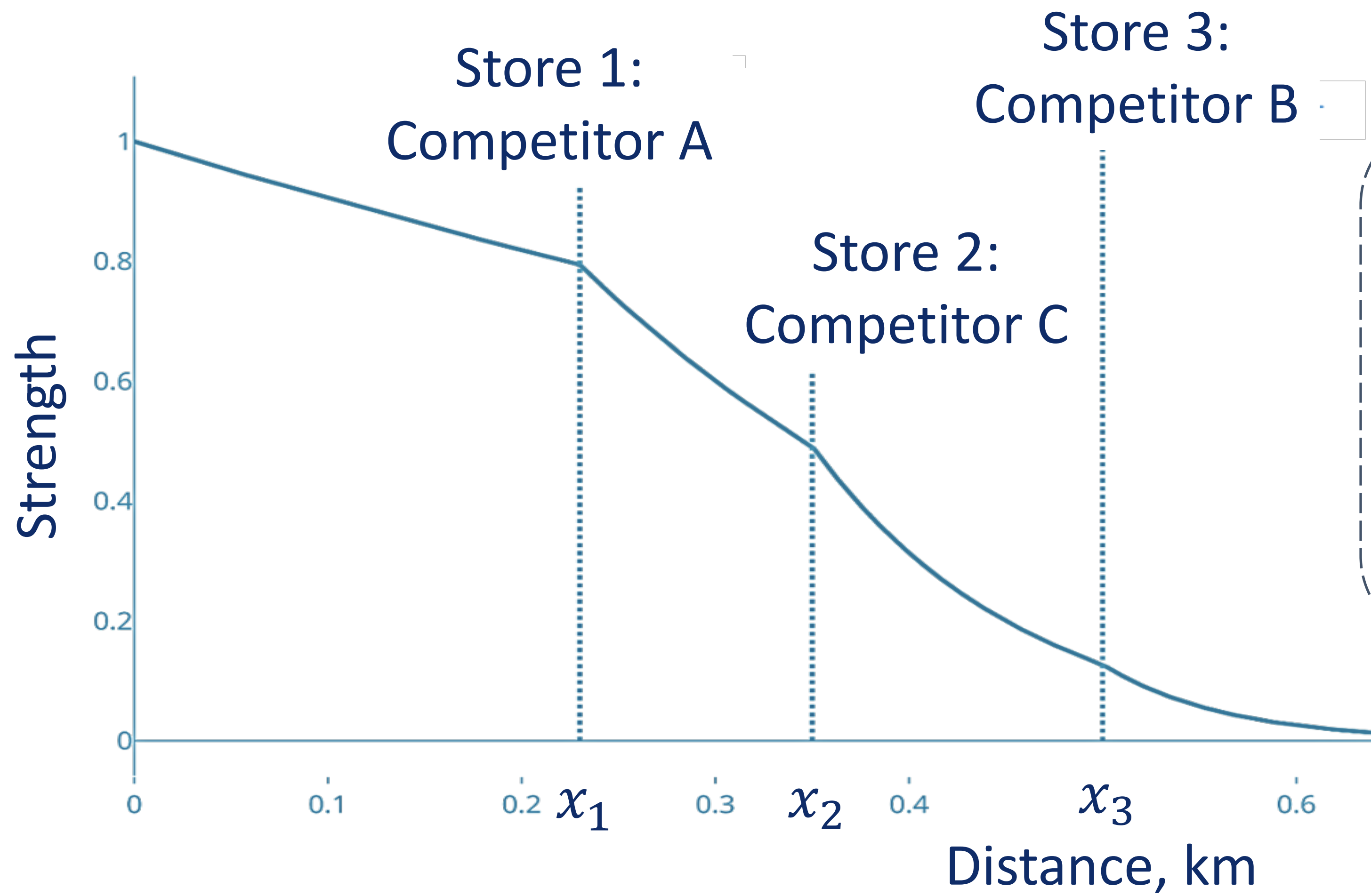
Method: 



Train set example:

Store	Store 1 Strength	Store 2 Strength	Store 3 Strength	…	Store 15 Strength
1	13	10	5		0
2	10	9	6		1
3	14	10	5		0
4	18	7	7		0
5	19	8	7		1
6	18	9	5		0

How to Calculate Competitor Strength



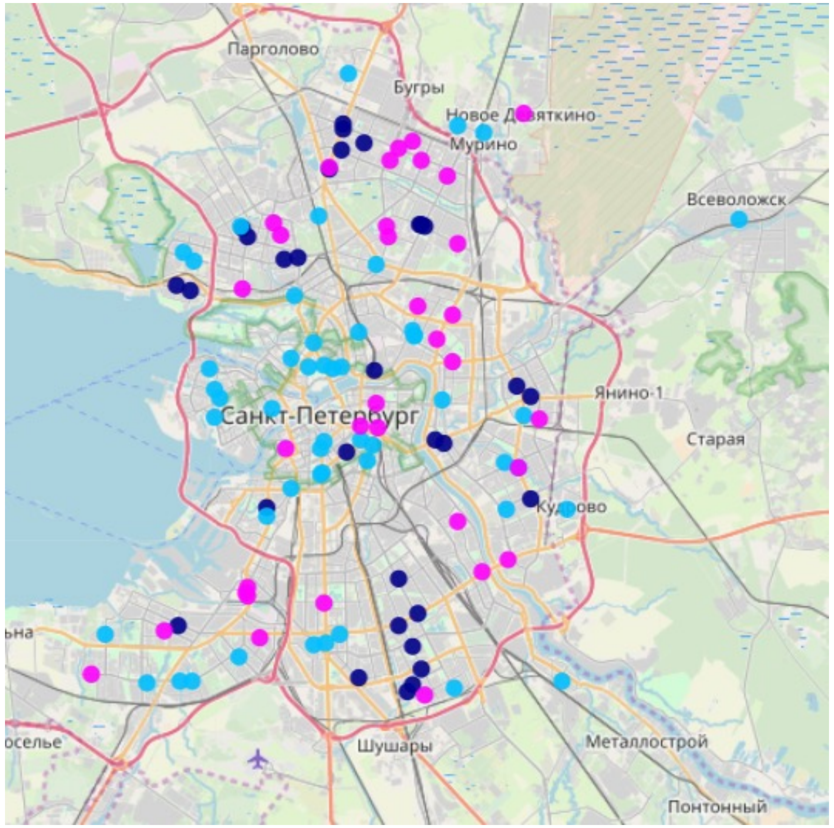
Business Heuristic:

$$k(x) = \begin{cases} e^{-x}, if\ x < x_1 \\ \vdots \\ c_n e^{-n^2 x}, if\ x_{n-1} < x < x_n \end{cases}$$

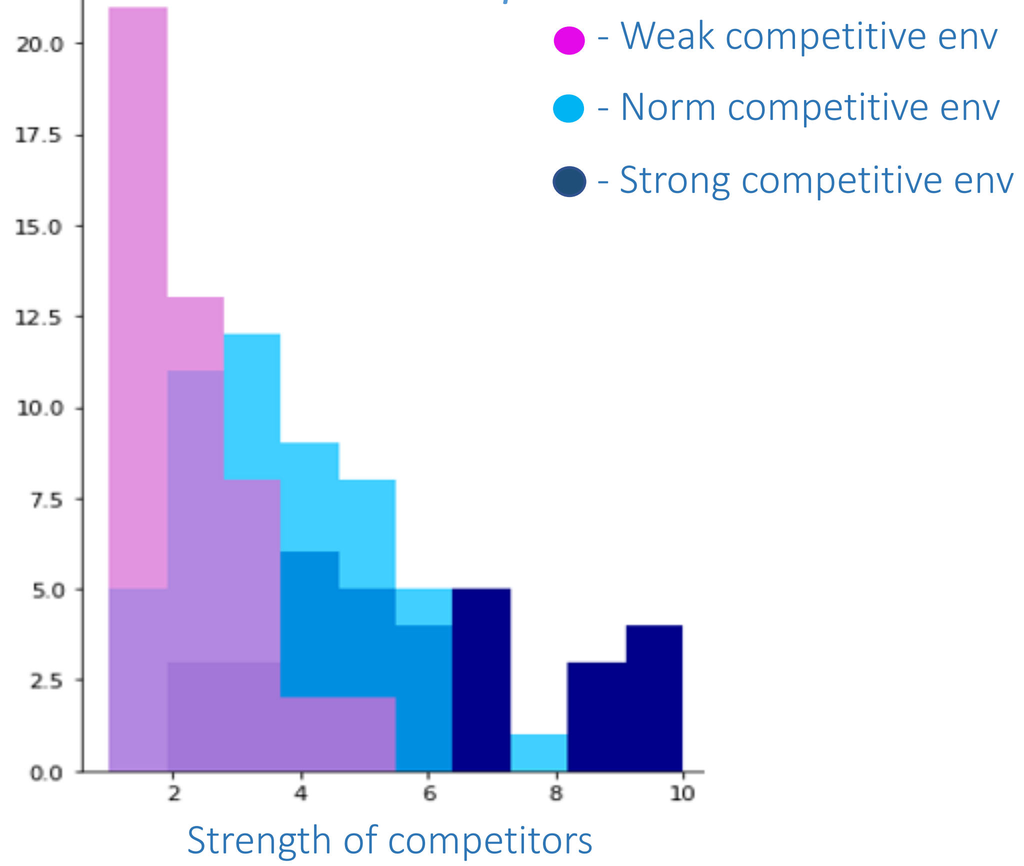
$$c_n = \exp\left\{\sum_{i=1}^{n-1} (2i+1)x_i\right\}$$

Results of Clusterization based on Competitive Environment

Clusterization Results

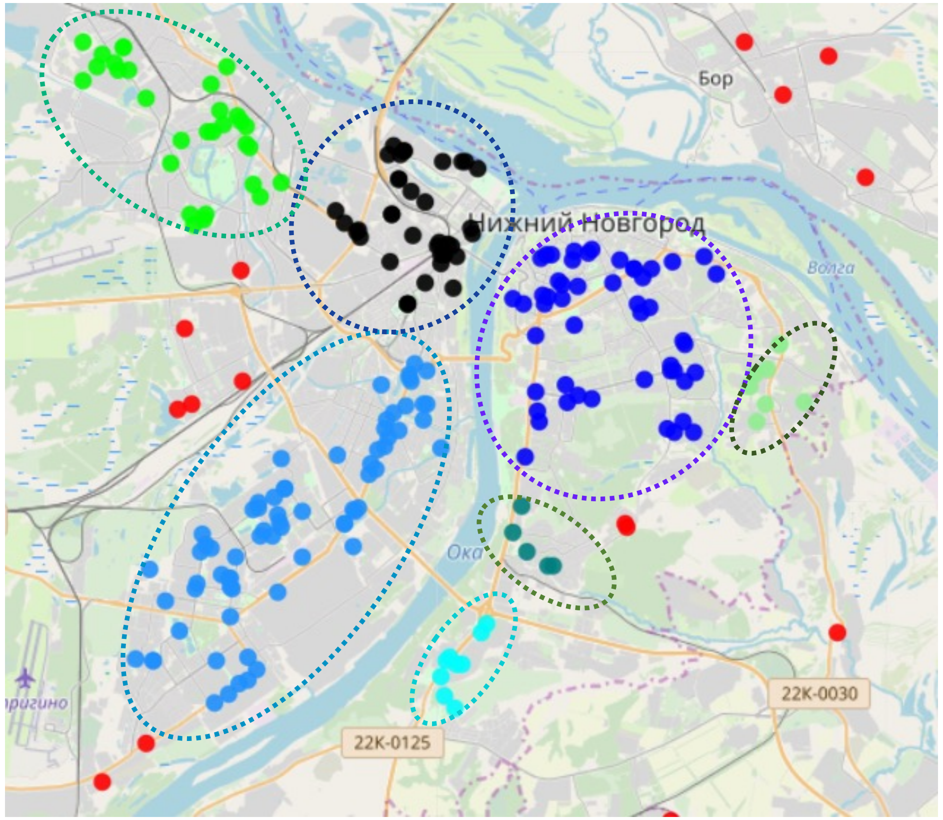


Cluster Description

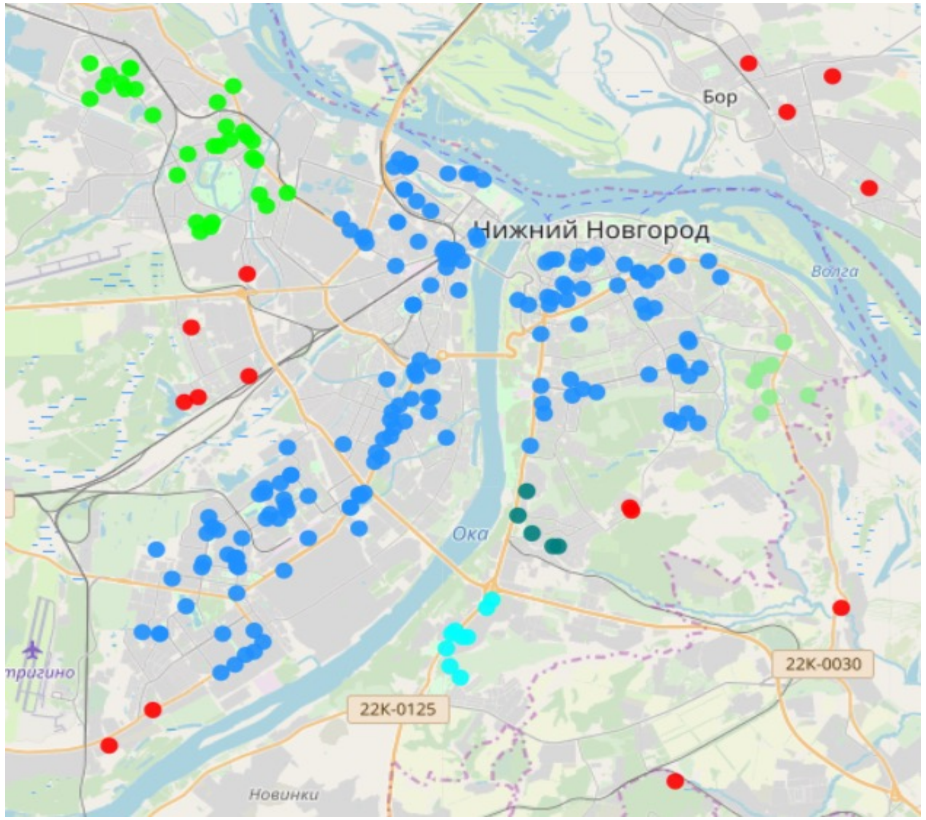


Clustering stores by distance using the example of a grocery chain.

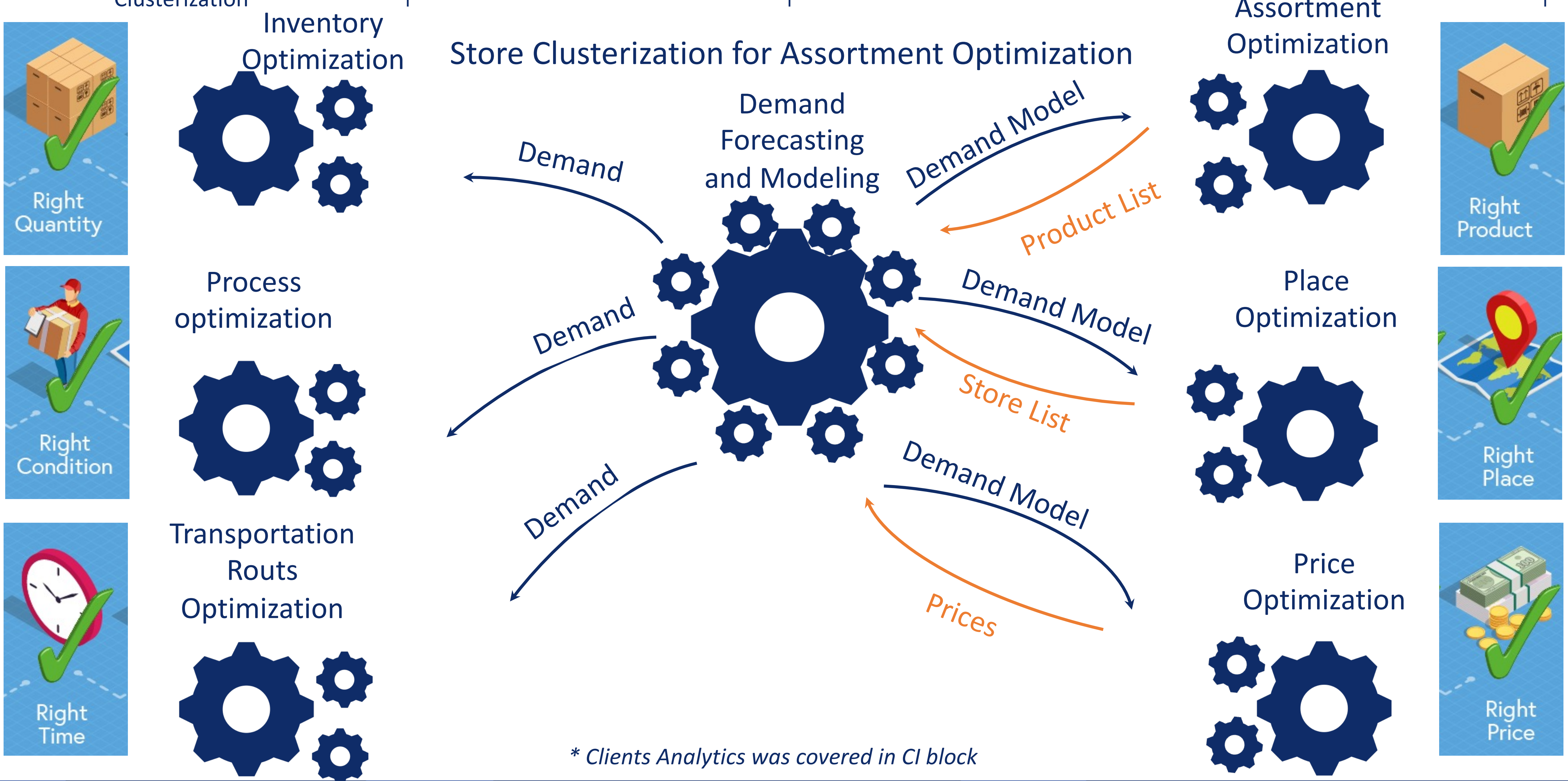
Pic. 1: ε = 15 min.



Pic. 2: ε = 25 min.



Store Clusterization for Assortment Optimization



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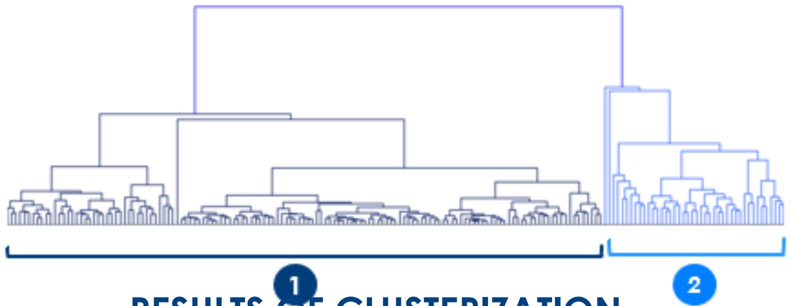
Requirements (Ideas) of the Store Clusterization for Assortment Optimization

Key drivers for Clusterization:

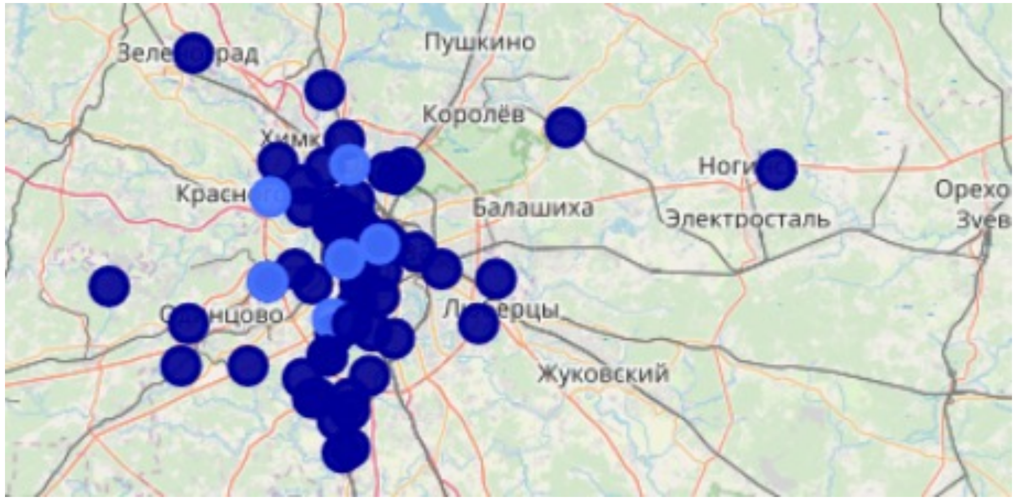
Prevailing categories in Sales History

Clusterization based on Sales Structure

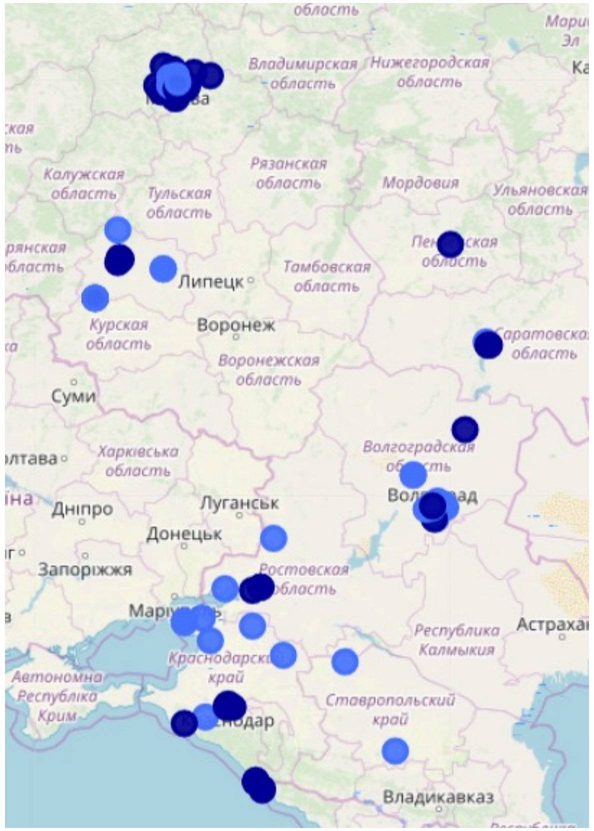
НА ДЕНДРОГРАММЕ МОЖНО ВЫДЕЛИТЬ 2 КЛАСТЕРА



RESULTS OF CLUSTERIZATION



RESULTS OF CLUSTERIZATION



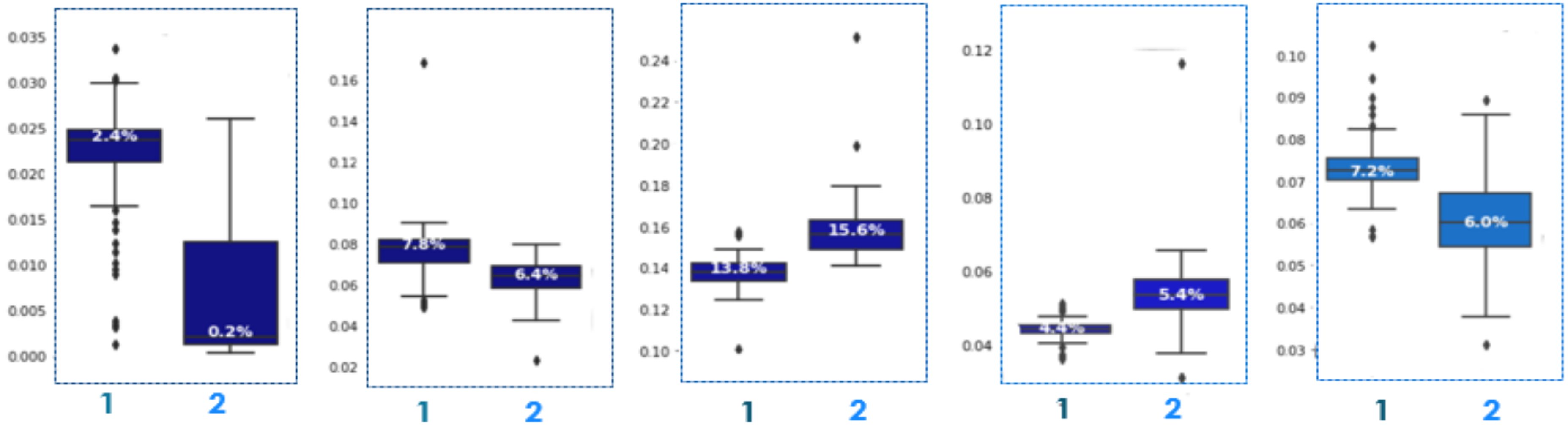
Hierarchical Clustering

Interpretation of Clusterization Results

PREVAILING CATEGORIES STATISTICS

Cluster	No.Stores	Square	Lifetime, years
1	137	95 (±32) M2	4.8 (±5)
2	42	59 (±17) M2	2.6 (±5)

Hierarchical Clustering



SKIN CARE CARDIOVASCULAR FLU MUSCULOSKELETAL VITAMINS

Agenda

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Defining KVI Baskets

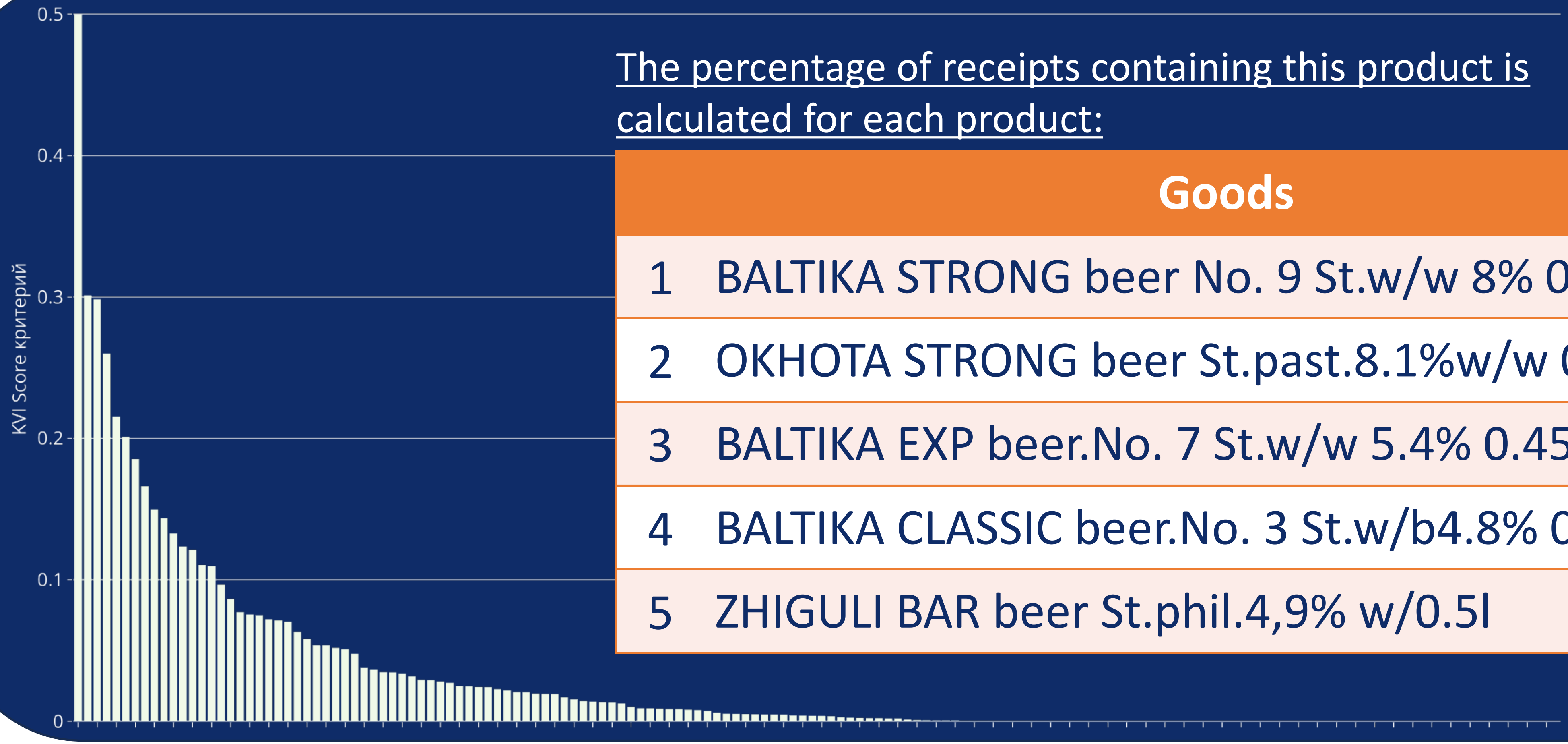
Setting the task:

1. Determine the list of KVI products by calculating the KVI “Score” criteria
2. Determine the optimal threshold for the KVI Score criterion

Input data:

1. Competitor price monitoring
2. Receipt data from January 1 to March 31, 2018
3. List of shopping centers and STMs

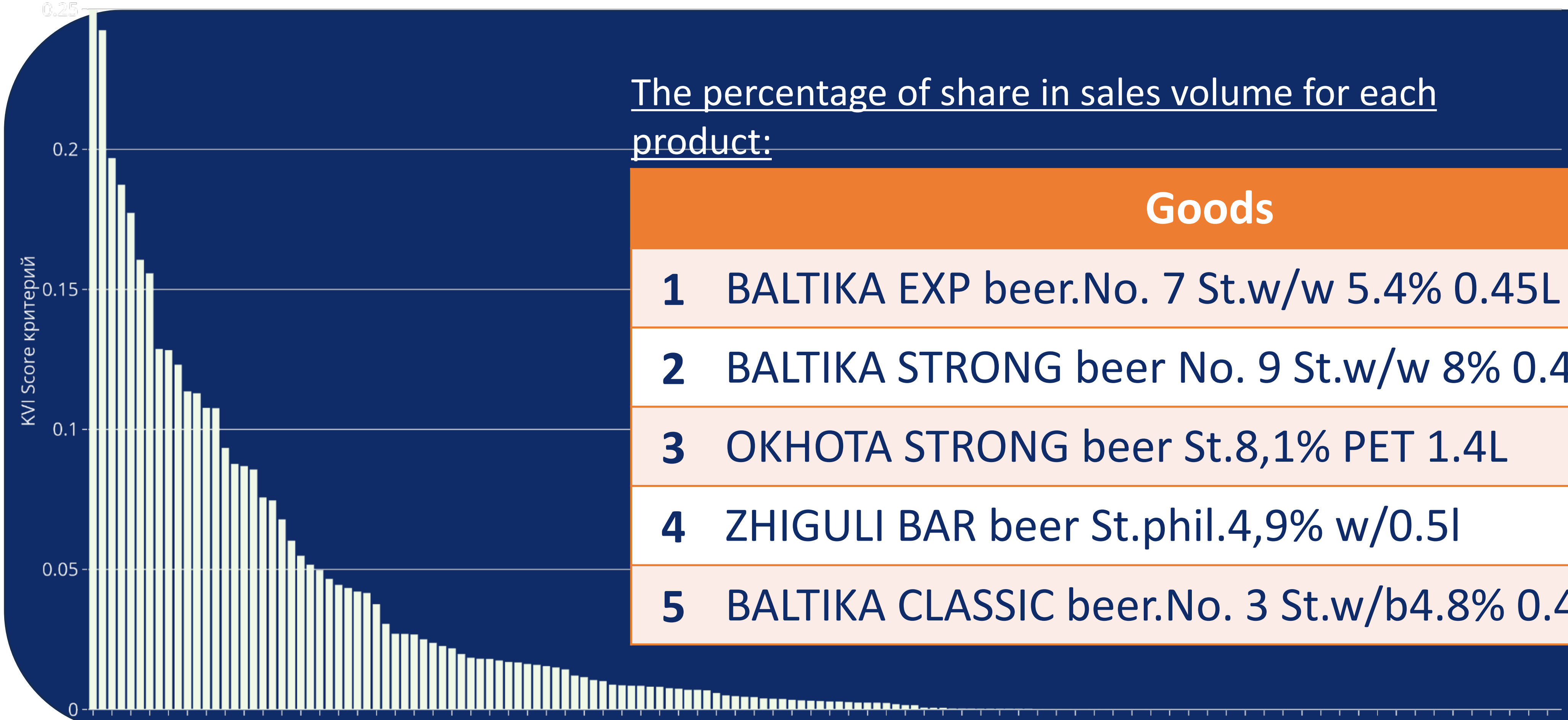
KVI Score criteria. Fitting into the cheque



The percentage of receipts containing this product is calculated for each product:

	Goods
1	BALTIKA STRONG beer No. 9 St.w/w 8% 0.45l
2	OKHOTA STRONG beer St.past.8.1%w/w 0.45l
3	BALTIKA EXP beer.No. 7 St.w/w 5.4% 0.45L
4	BALTIKA CLASSIC beer.No. 3 St.w/b4.8% 0.45L
5	ZHIGULI BAR beer St.phil.4,9% w/0.5l

KVI Score criteria. Share in sales volume



The percentage of share in sales volume for each product:

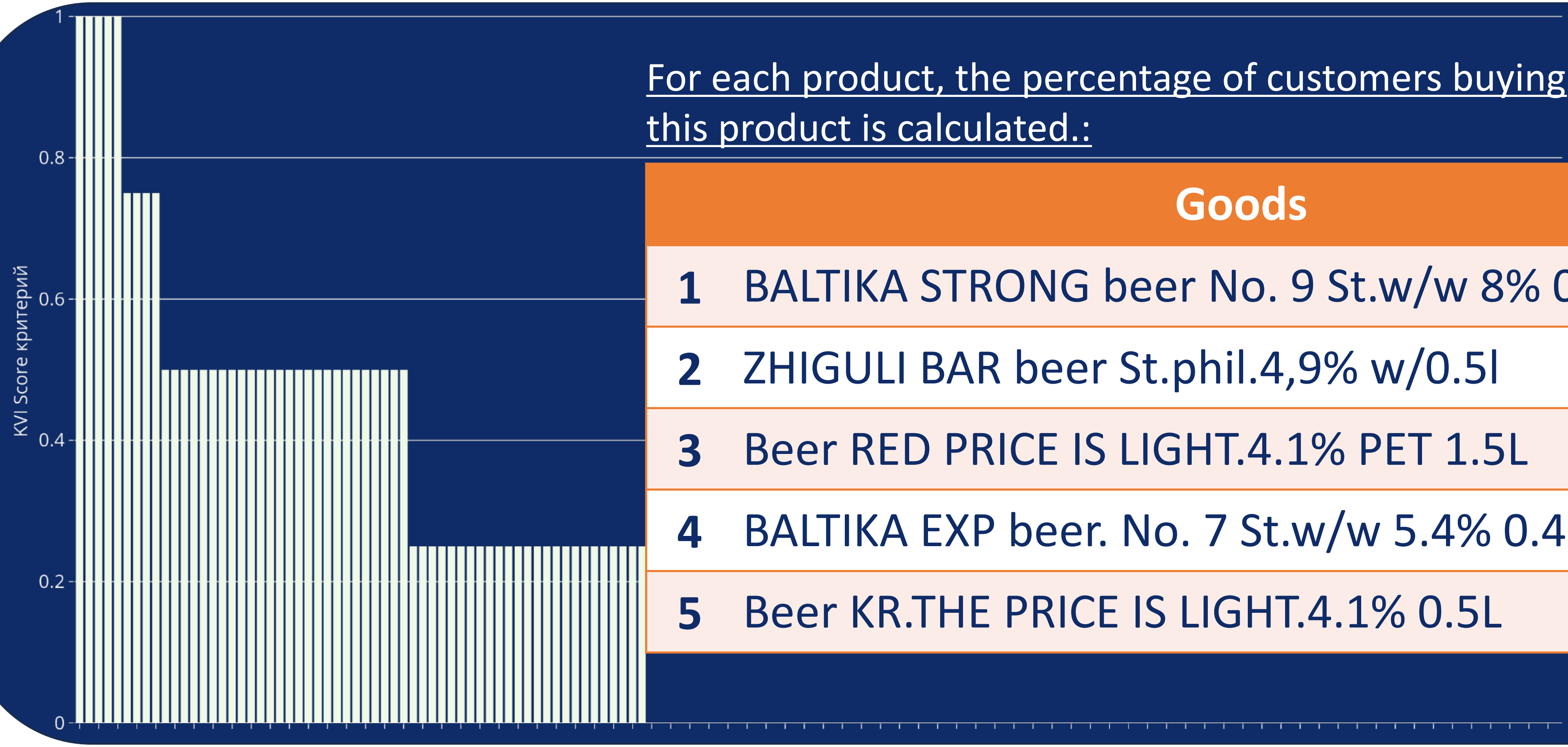
	Goods
1	BALTIKA EXP beer.No. 7 St.w/w 5.4% 0.45L
2	BALTIKA STRONG beer No. 9 St.w/w 8% 0.45l
3	OKHOTA STRONG beer St.8,1% PET 1.4L
4	ZHIGULI BAR beer St.phil.4,9% w/0.5l
5	BALTIKA CLASSIC beer.No. 3 St.w/b4.8% 0.45L

KVI Score criteria. The criterion of the share of buyers

For each product, the percentage of customers buying this product is calculated.:

	Goods
1	BALTIKA EXP beer.No. 7 St.w/w 5.4% 0.45L
2	ZHIGULI BAR beer St.phil.4,9% w/0.5l
3	BALTIKA CLASSIC beer.No. 3 St.w/b4.8% 0.45L
4	Beer ZOL.BARREL CLASS.sv.5,2% w/w 0.45l
5	BALTIKA STRONG beer No. 9 St.w/w 8% 0.45l

KVI Score criteria. Criteria for the level of competition



<u>For each product, the percentage of customers buying this product is calculated.:</u>

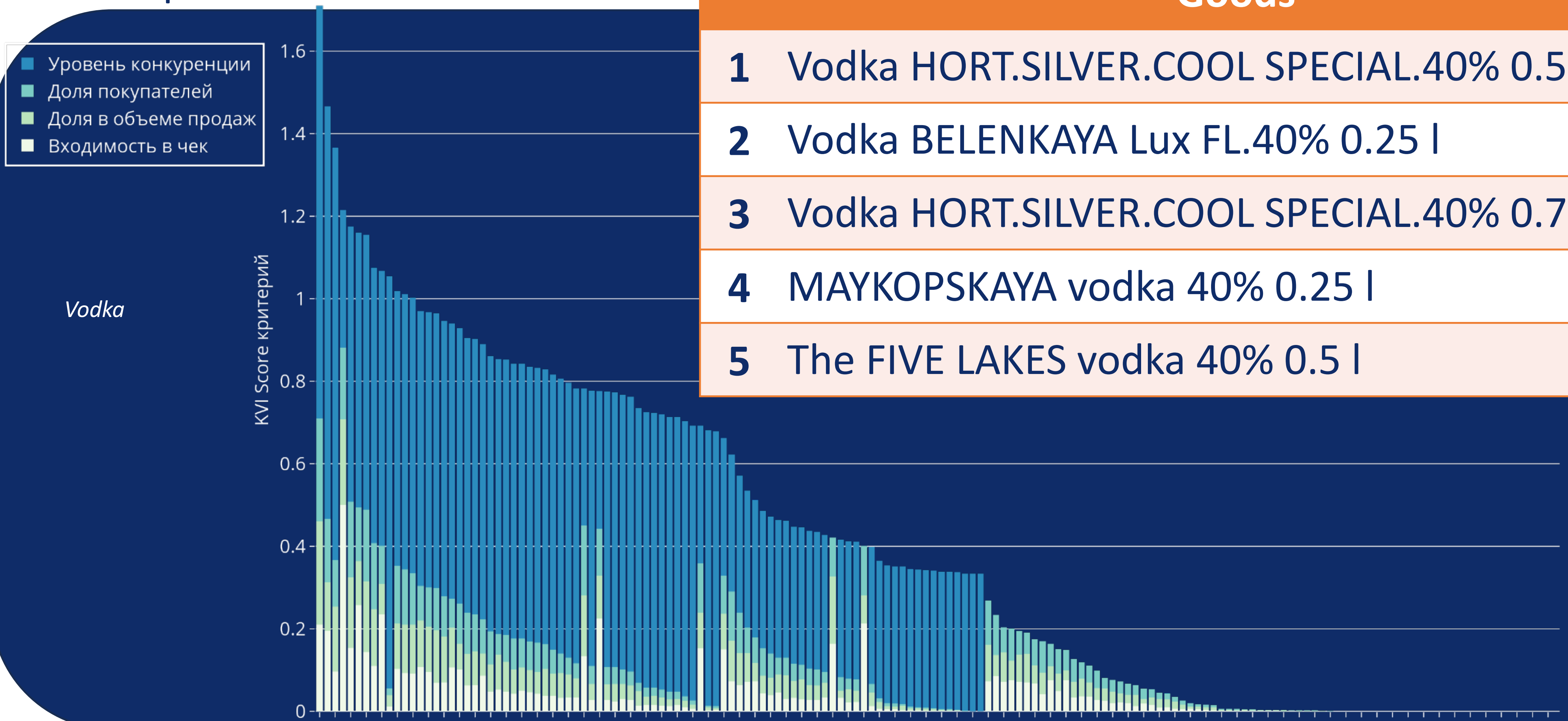
	Goods
1	BALTIKA STRONG beer No. 9 St.w/w 8% 0.45l
2	ZHIGULI BAR beer St.phil.4,9% w/0.5l
3	Beer RED PRICE IS LIGHT.4.1% PET 1.5L
4	BALTIKA EXP beer. No. 7 St.w/w 5.4% 0.45L
5	Beer KR.THE PRICE IS LIGHT.4.1% 0.5L

KVI Score criteria. A single KVI Score criterion

Goods	
1	STRONG beer BALTIKA №9 St W/b 8% 0.45 l
2	Beer BALTIKA EXPORT No. 7 St W/b 5,4% 0.45 l
3	Beer ZHIGULI barnoe St. Phil.4,9% W/W 0.5 l
4	Beer RED PRICE IS LIGHT.4.1% of PET 1.5 l
5	ZHIGULI BAR beer St.phil.4,9% w/0.5l

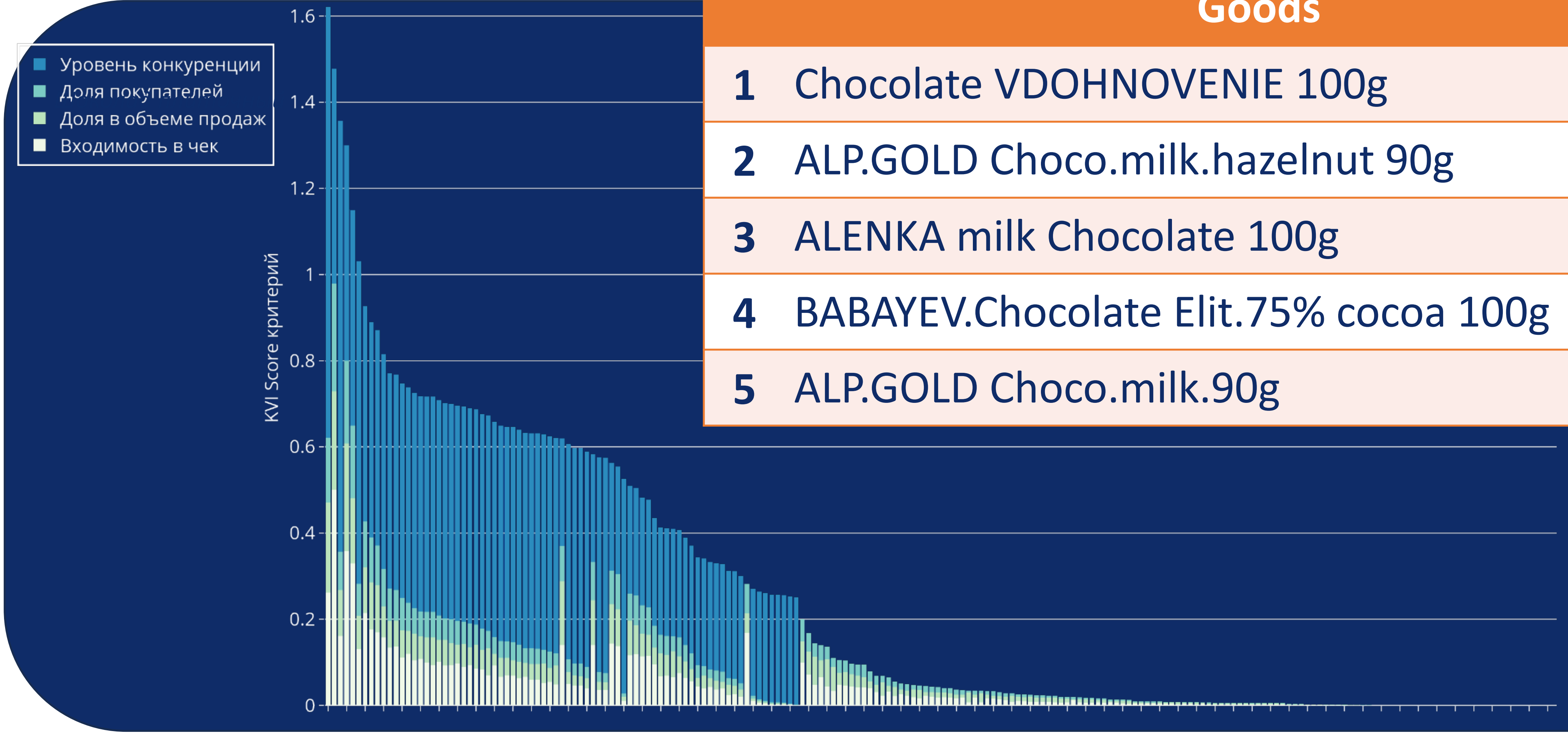
Examples

Vodka



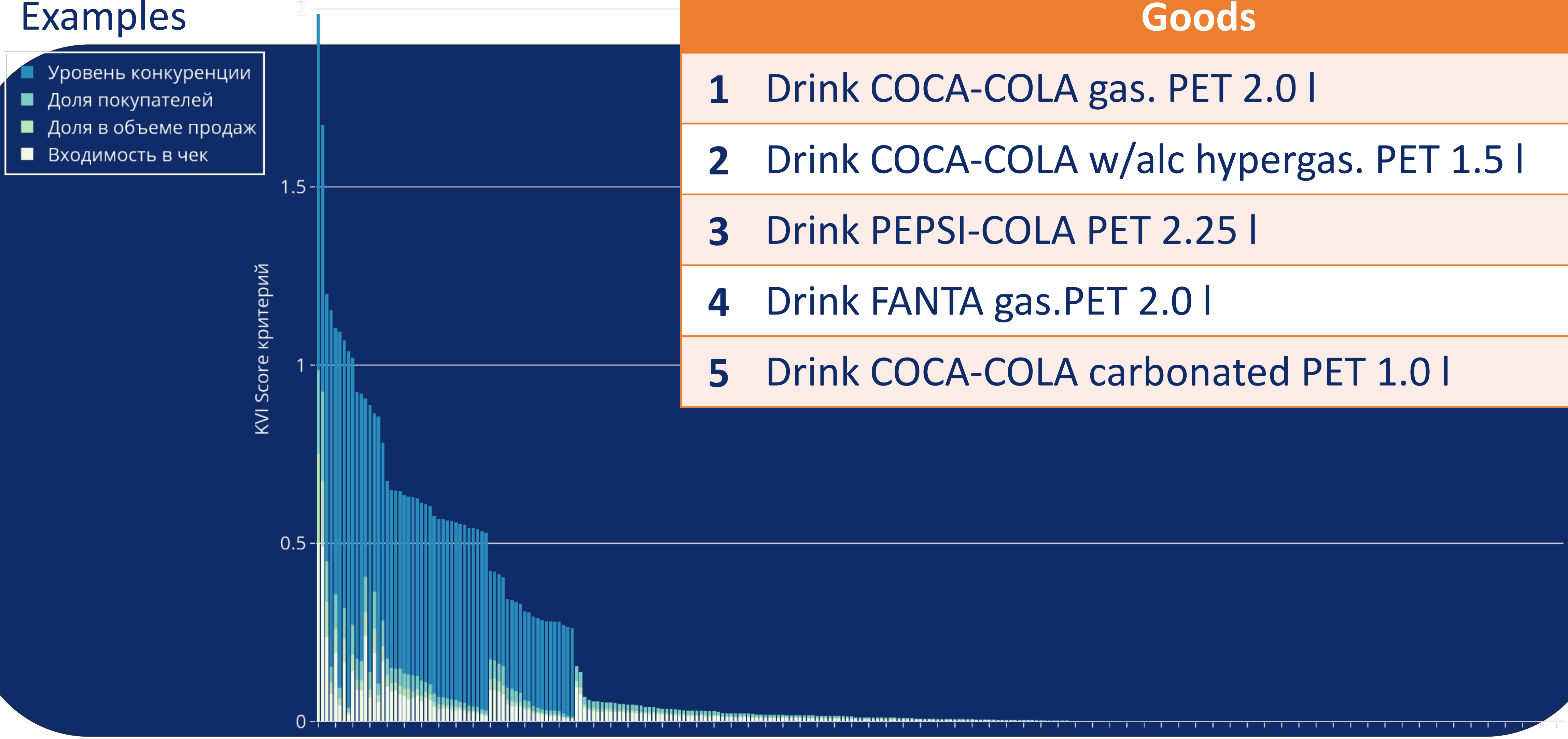
	Goods
1	Vodka HORT.SILVER.COOL SPECIAL.40% 0.5 l
2	Vodka BELENKAYA Lux FL.40% 0.25 l
3	Vodka HORT.SILVER.COOL SPECIAL.40% 0.7 l
4	MAYKOPSKAYA vodka 40% 0.25 l
5	The FIVE LAKES vodka 40% 0.5 l

Examples



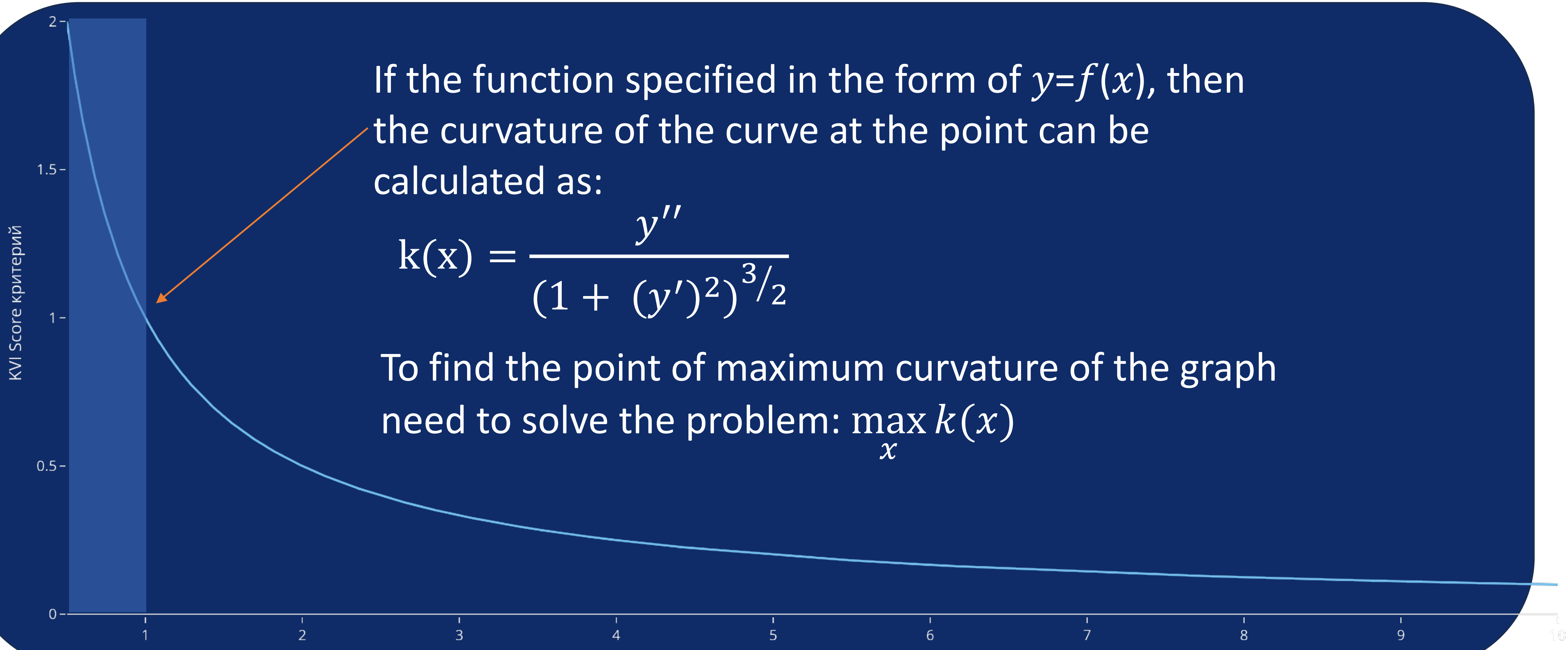
	Goods
1	Chocolate VDOHNOVENIE 100g
2	ALP.GOLD Choco.milk.hazelnut 90g
3	ALENKA milk Chocolate 100g
4	BABAYEV.Chocolate Elit.75% cocoa 100g
5	ALP.GOLD Choco.milk.90g

Examples



	Goods
1	Drink COCA-COLA gas. PET 2.0 l
2	Drink COCA-COLA w/alc hypergas. PET 1.5 l
3	Drink PEPSI-COLA PET 2.25 l
4	Drink FANTA gas.PET 2.0 l
5	Drink COCA-COLA carbonated PET 1.0 l

The algorithm for determining the optimal threshold

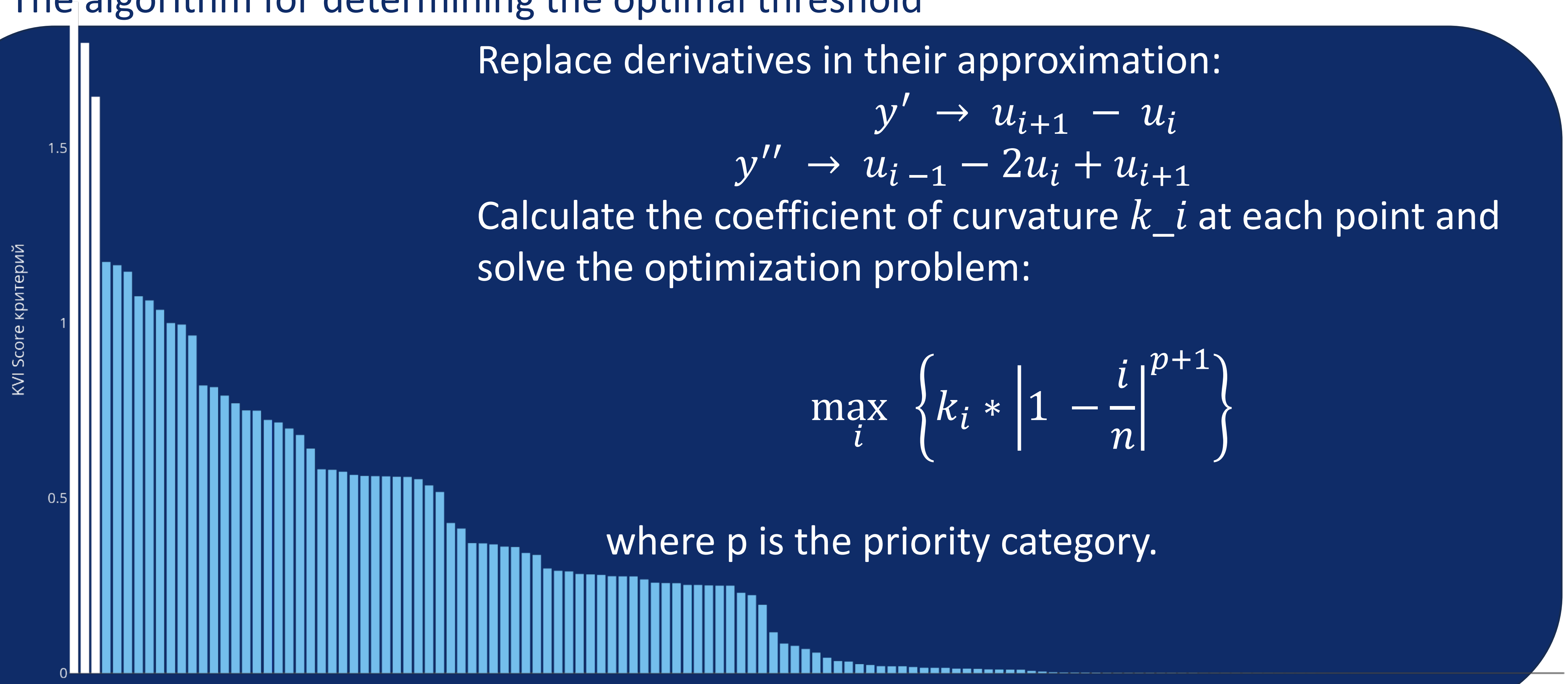


If the function specified in the form of $y=f(x)$, then the curvature of the curve at the point can be calculated as:

$$k(x) = \frac{y''}{(1 + (y')^2)^{3/2}}$$

To find the point of maximum curvature of the graph need to solve the problem: $\max\limits_{x} k(x)$

The algorithm for determining the optimal threshold



Replace derivatives in their approximation:

$$y' \rightarrow u_{i+1} - u_i$$

$$y'' \rightarrow u_{i-1} - 2u_i + u_{i+1}$$

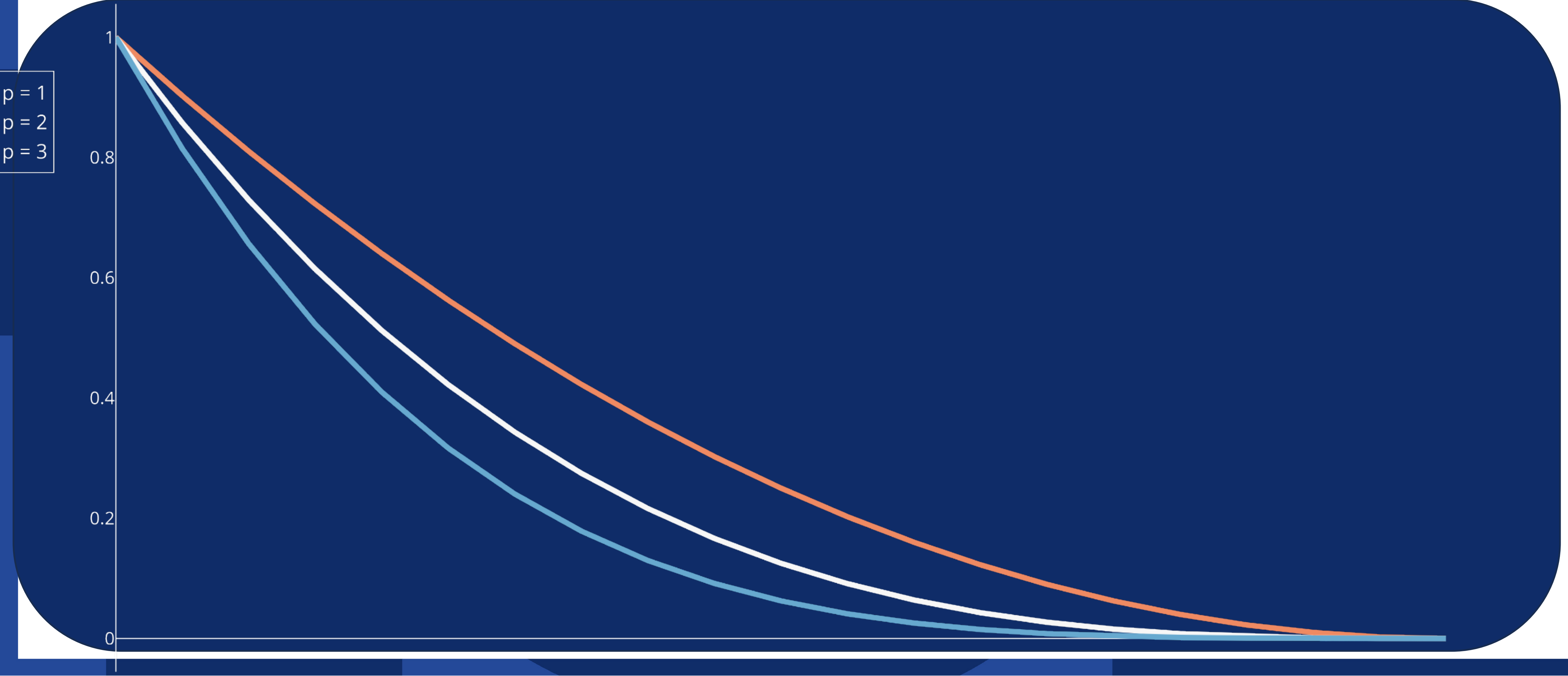
Calculate the coefficient of curvature k_i at each point and solve the optimization problem:

$$\max_i \left\{ k_i * \left|1 - \frac{i}{n}\right|^{p+1} \right\}$$

where p is the priority category.

Prioritizing category

Виды штрафующей функции



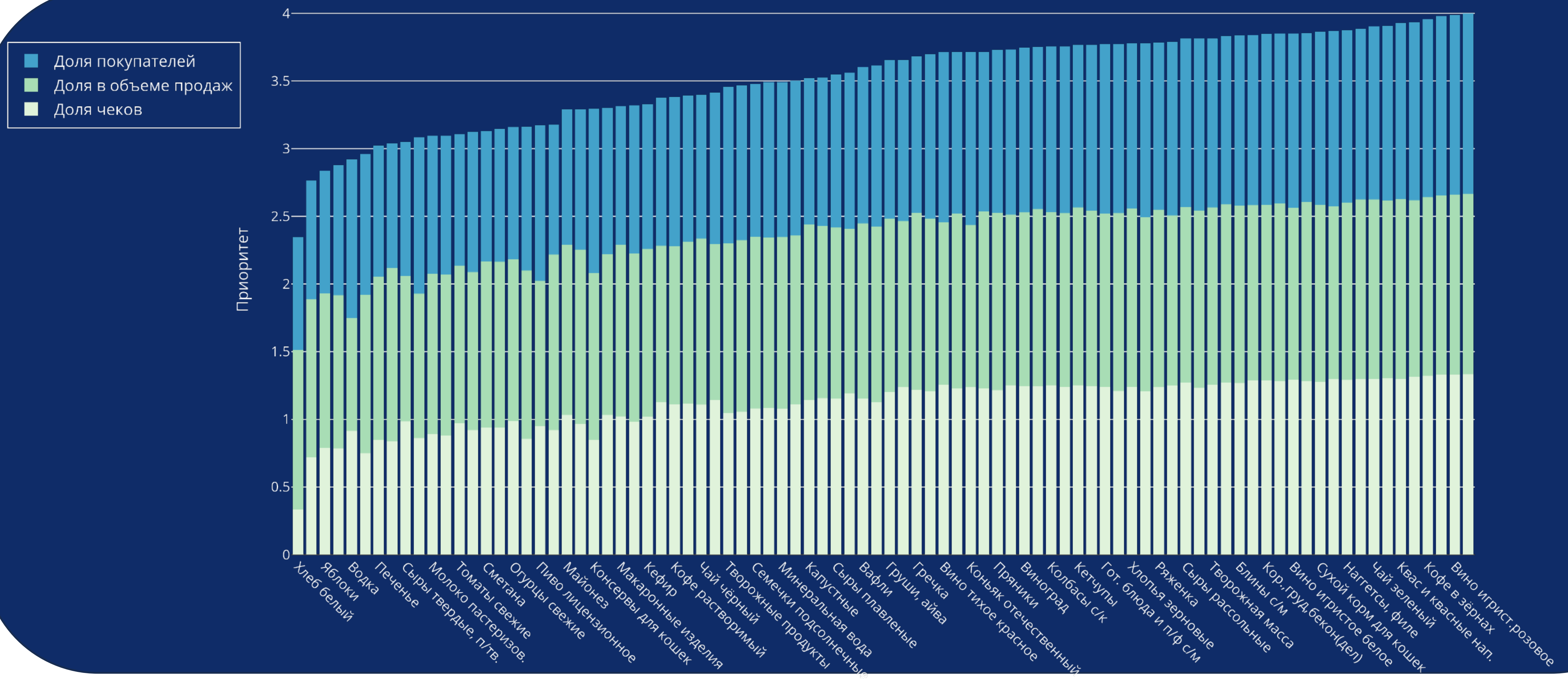
Prioritizing category

For each category of goods is calculated features:

1. Share of receipts
2. Share in sales volume
3. Customer share

Based on them, the priority of the p ∈ category is determined [1;3]

Prioritizing category

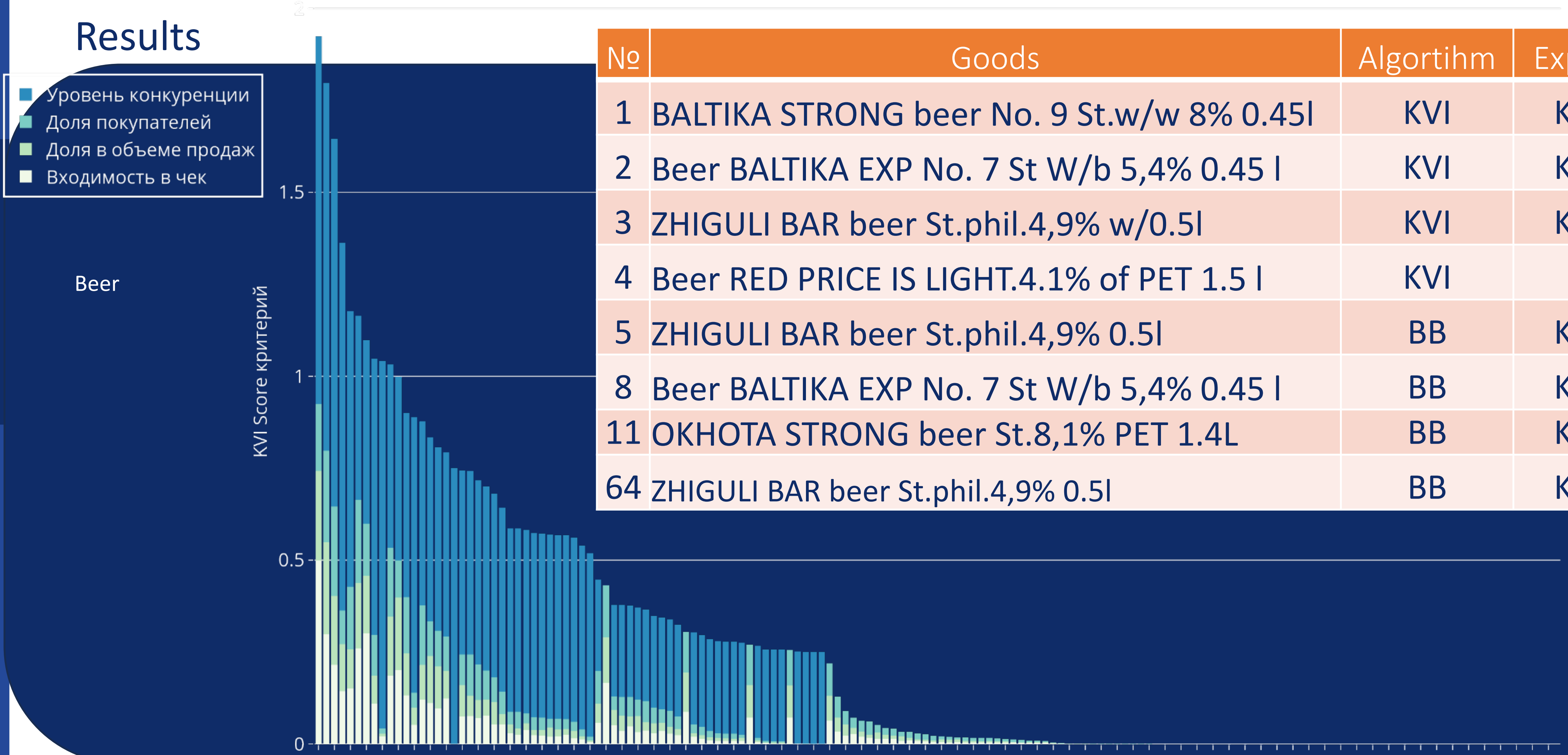


The algorithm for determining the optimal threshold

№	Goods	Algortihm	Expert
1	BALTIKA STRONG beer No. 9 St.w/w 8% 0.45l	KVI	KVI
2	Beer BALTIKA EXP No. 7 St W/b 5,4% 0.45 l	KVI	KVI
3	ZHIGULI BAR beer St.phil.4,9% w/0.5l	KVI	KVI
4	Beer RED PRICE IS LIGHT.4.1% of PET 1.5 l	KVI	
5	ZHIGULI BAR beer St.phil.4,9% 0.5l	BB	KVI
8	Beer BALTIKA EXP No. 7 St W/b 5,4% 0.45 l	BB	KVI
11	OKHOTA STRONG beer St.8,1% PET 1.4L	BB	KVI
64	ZHIGULI BAR beer St.phil.4,9% 0.5l	BB	KVI

1. Find the point of maximum curvature of the graph
2. KVI left, right, Back Basket

Results



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64	ZHIGULI BAR beer St.phil.4,9% 0.5l	BB	KVI

Results

№	Goods	Algortihm	Expert
1	Chocolate VDOHNOVENIE 100g	KVI	BB
2	ALP.GOLD Choko.milk.hazelnut 90g	KVI	BB
3	ALENKA milk Chocolate 100g	BB	KVI
6	ALENKA A LOT OF MILK Chocolate 100g	BB	KVI
57	Choko.KINDER CHOC.mol.w.mol.fill. 100g	BB	KVI

Results

Chocolate tile

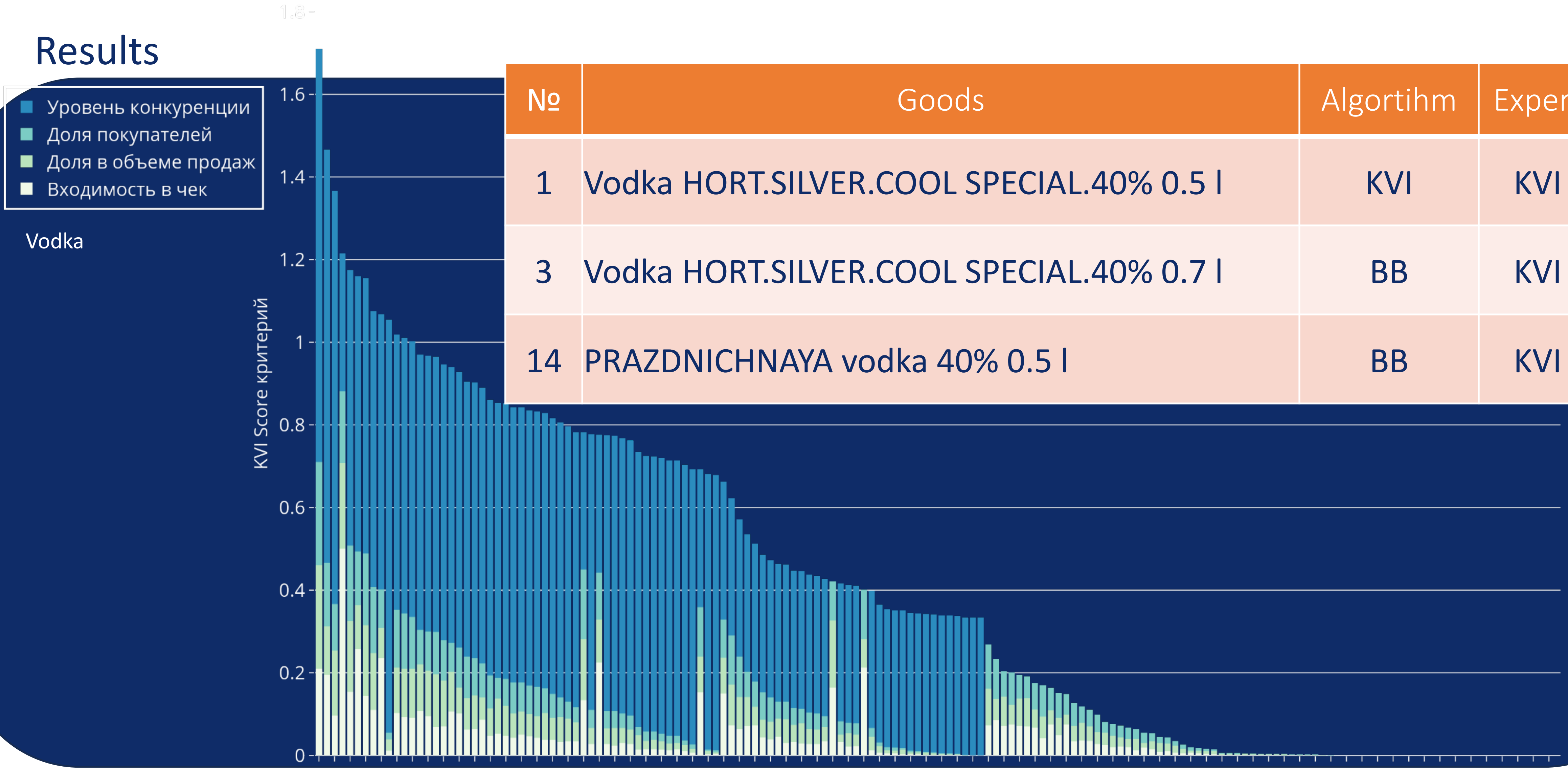
№	Goods	Algortihm	Expert
1	Chocolate VDOHNOVENIE 100g	KVI	BB
2	ALP.GOLD Choko.milk.hazelnut 90g	KVI	BB
3	ALENKA milk Chocolate 100g	BB	KVI
6	ALENKA A LOT OF MILK Chocolate 100g	BB	KVI
57	Choko.KINDER CHOC.mol.w.mol.fill. 100g	BB	KVI

Results

№	Goods	Algortihm	Expert
1	Vodka HORT.SILVER.COOL SPECIAL.40% 0.5 l	KVI	KVI
3	Vodka HORT.SILVER.COOL SPECIAL.40% 0.7 l	BB	KVI
14	PRAZDNICHNAYA vodka 40% 0.5 l	BB	KVI

Results

Vodka



№	Goods	Algortihm	Expert
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14	PRAZDNICHNAYA vodka 40% 0.5 l	BB	KVI

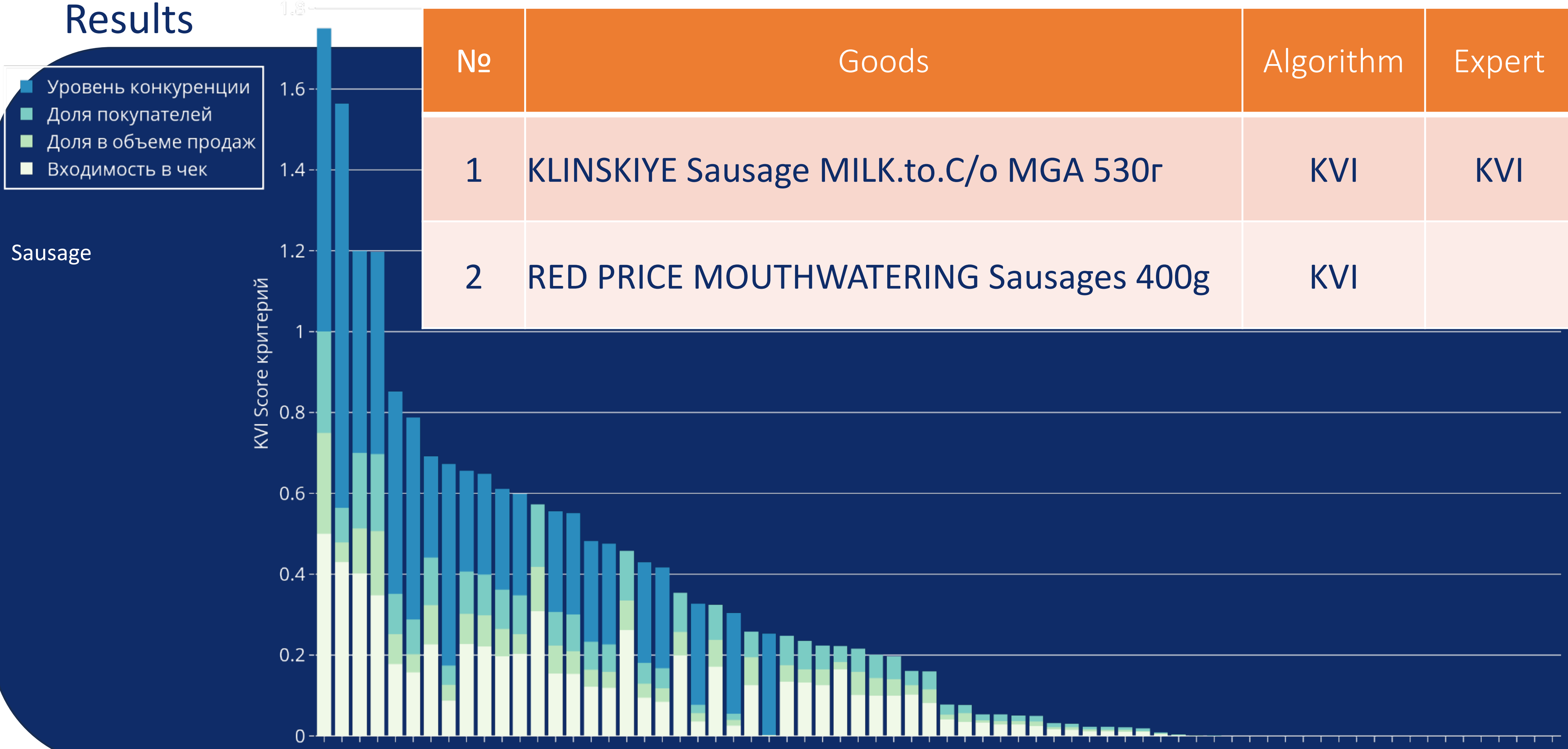
Results

№	Goods	Algorithm	Expert
1	KLINSKIYE Sausage MILK.to.C/o MGA 530г	KVI	KVI
2	RED PRICE MOUTHWATERING Sausages 400g	KVI	

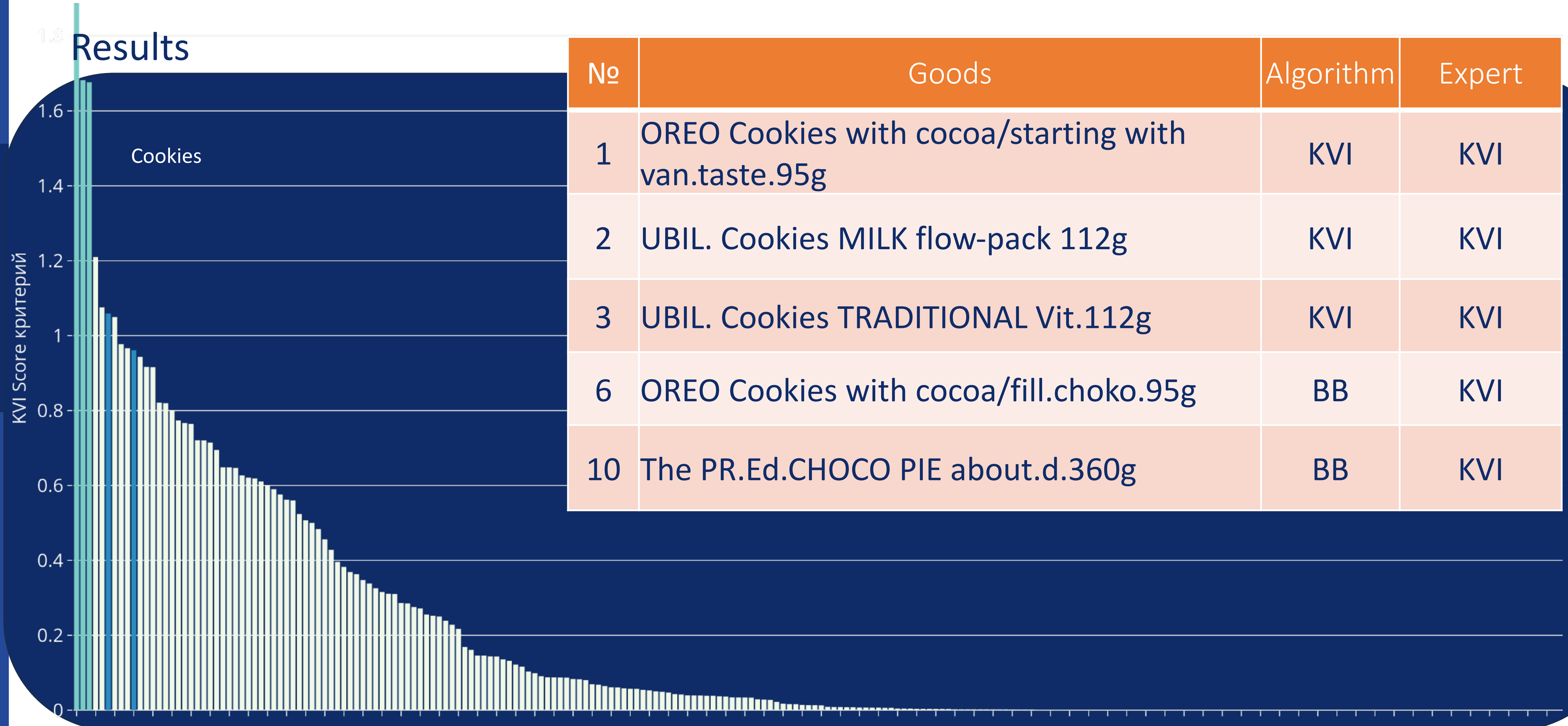
Results

№	Goods	Algorithm	Expert
1	KLINSKIYE Sausage MILK.to.C/o MGA 530г	KVI	KVI
2	RED PRICE MOUTHWATERING Sausages 400g	KVI	

Sausage

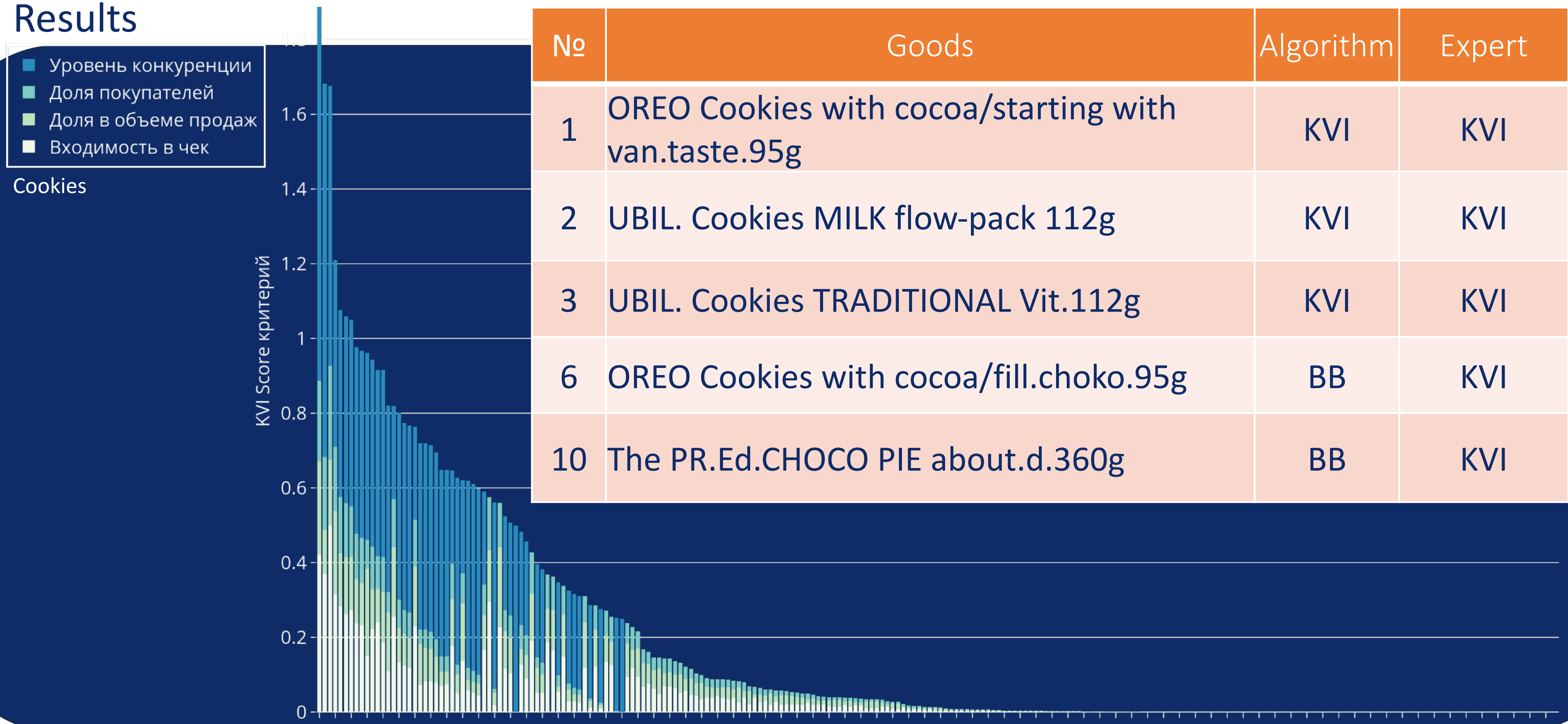


Results



№	Goods	Algorithm	Expert
1	OREO Cookies with cocoa/starting with van.taste.95g	KVI	KVI
2	UBIL. Cookies MILK flow-pack 112g	KVI	KVI
3	UBIL. Cookies TRADITIONAL Vit.112g	KVI	KVI
6	OREO Cookies with cocoa/fill.choko.95g	BB	KVI
10	The PR.Ed.CHOCO PIE about.d.360g	BB	KVI

Results



№	Goods	Algorithm	Expert
1	OREO Cookies with cocoa/starting with van.taste.95g	KVI	KVI
2	UBIL. Cookies MILK flow-pack 112g	KVI	KVI
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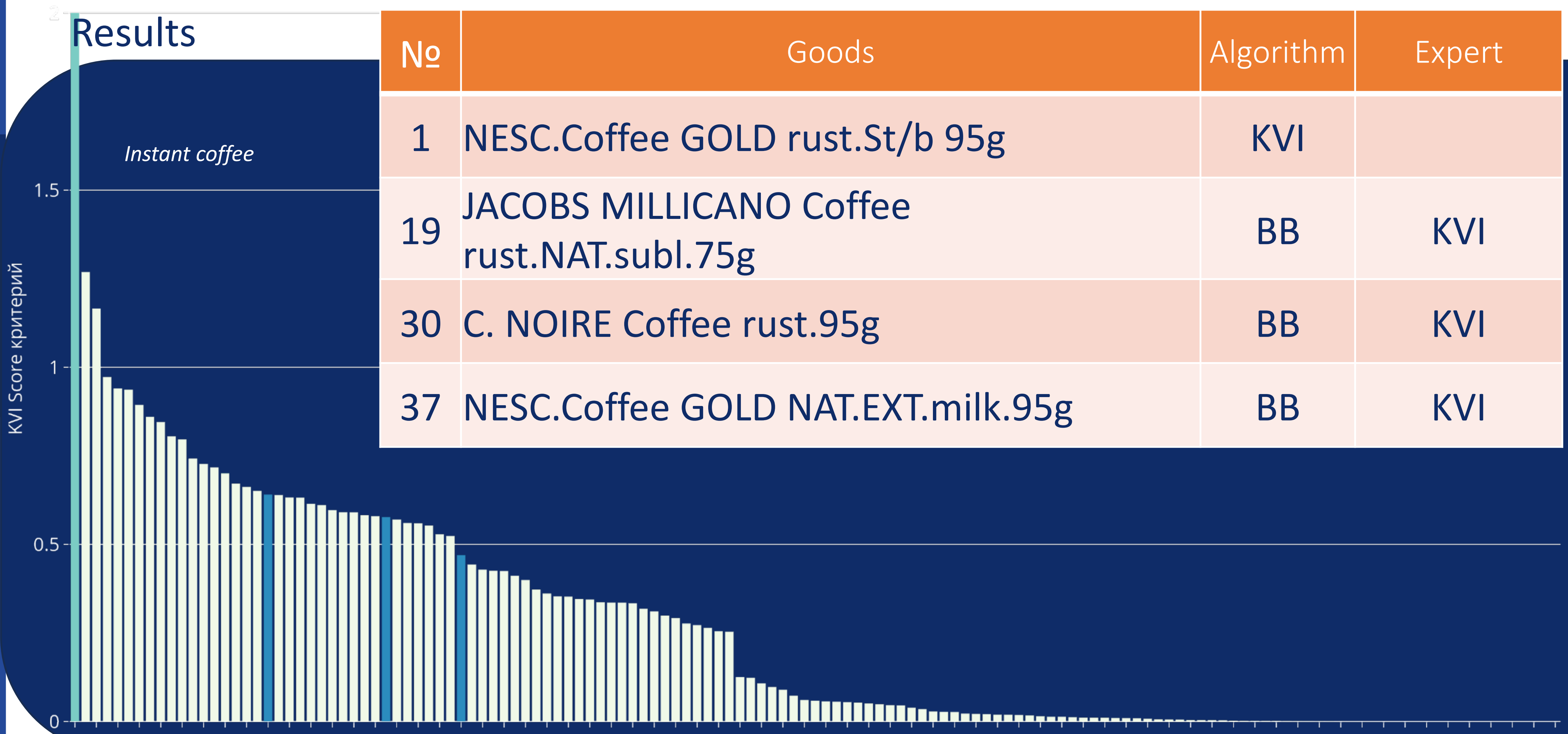
Results

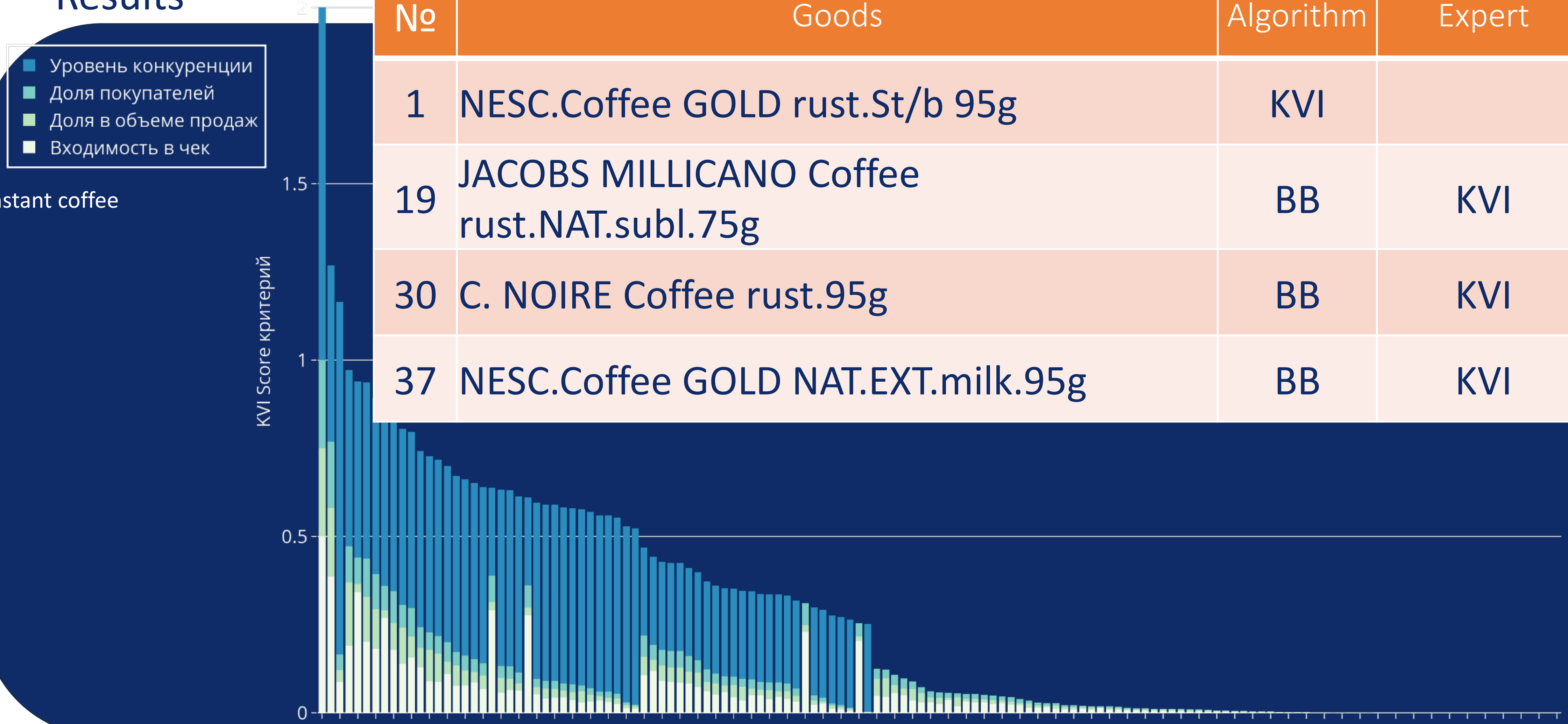


№	Goods	Algorithm	Expert
1	NESC.Coffee GOLD rust.St/b 95g	KVI	
19	JACOBS MILLICANO Coffee rust.NAT.subl.75g	BB	KVI
30	C. NOIRE Coffee rust.95g	BB	KVI
37	NESC.Coffee GOLD NAT.EXT.milk.95g	BB	KVI

Results

№	Goods	Algorithm	Expert
1	NESC.Coffee GOLD rust.St/b 95g	KVI	
19	JACOBS MILLICANO Coffee rust.NAT.subl.75g	BB	KVI
30	C. NOIRE Coffee rust.95g	BB	KVI
37	NESC.Coffee GOLD NAT.EXT.milk.95g	BB	KVI

Instant coffee

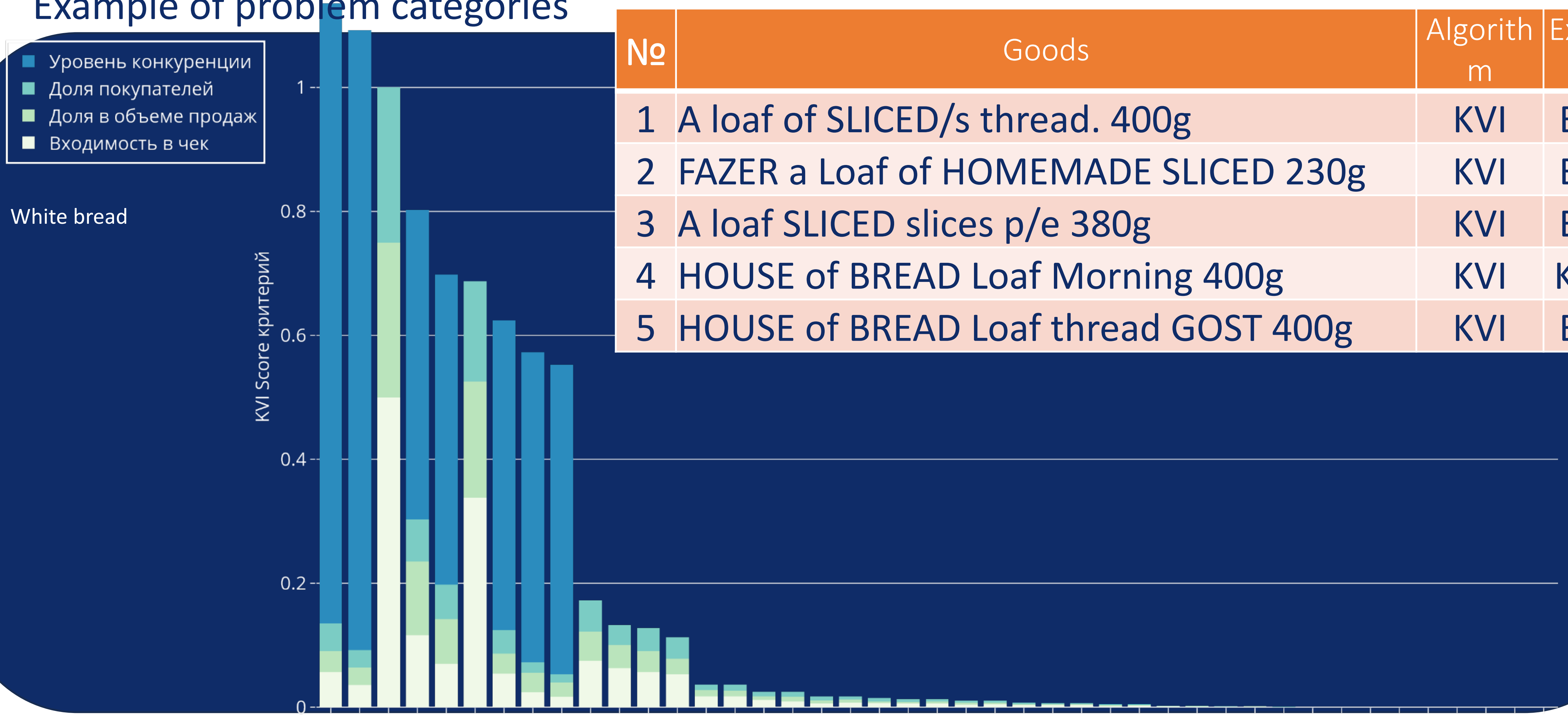


Problematic categories

After determining the optimal threshold partitioning is checked: "KVI products < 10% of the number of items in a category".
If the algorithm has identified > 10%, this category falls within the report.

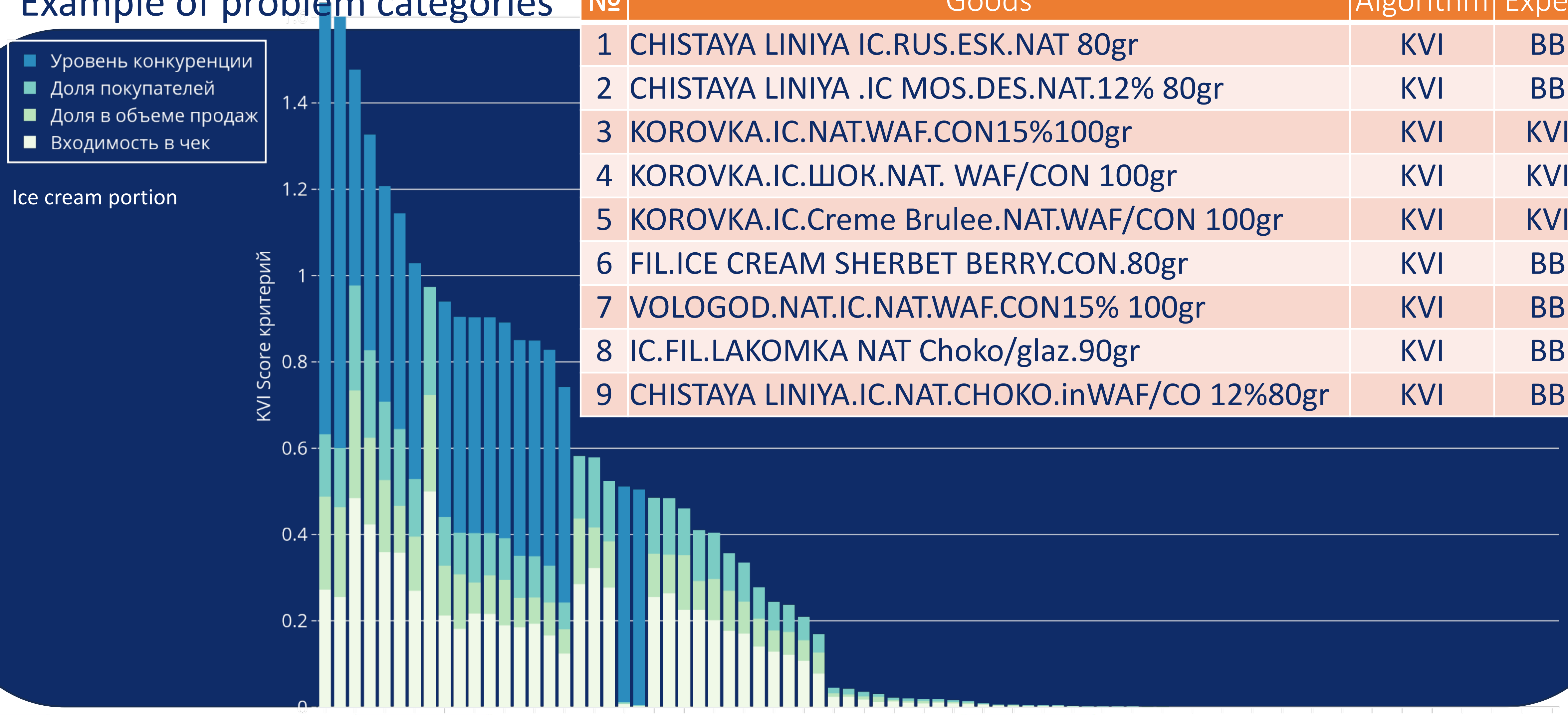
Category	Algorithm	Expert
Crab. Sticks, meat	2	2
Ice cream portions	9	3
Fresh Cucumbers	2	1
Spaghetti	2	2
White bread	5	1

Example of problem categories



№	Goods	Algorithm	Expert
1	A loaf of SLICED/s thread. 400g	KVI	BB
2	FAZER a Loaf of HOMEMADE SLICED 230g	KVI	BB
3	A loaf SLICED slices p/e 380g	KVI	BB
4	HOUSE of BREAD Loaf Morning 400g	KVI	KVI
5	HOUSE of BREAD Loaf thread GOST 400g	KVI	BB

Example of problem categories



№	Goods	Algorithm	Expert
1	CHISTAYA LINIYA IC.RUS.ESK.NAT 80gr	KVI	BB
2	CHISTAYA LINIYA .IC MOS.DES.NAT.12% 80gr	KVI	BB
3	KOROVKA.IC.NAT.WAF.CON15%100gr	KVI	KVI
4	KOROVKA.IC.ШОК.NAT. WAF/CON 100gr	KVI	KVI
5	KOROVKA.IC.Creme Brulee.NAT.WAF/CON 100gr	KVI	KVI
6	FIL.ICE CREAM SHERBET BERRY.CON.80gr	KVI	BB
7	VOLOGOD.NAT.IC.NAT.WAF.CON15% 100gr	KVI	BB
8	IC.FIL.LAKOMKA NAT Choko/glaz.90gr	KVI	BB
9	CHISTAYA LINIYA.IC.NAT.CHOKO.inWAF/CO 12%80gr	KVI	BB

Agenda

- Forecast Accuracy Measurement
- Store Clusterization in Retail
- Building KVI products
- HW #3
    - Demand Forecasting

Contest: https://www.kaggle.com/competitions/dscs-winter24-hw3/host

Forecast Horizon: 4 weeks ahead

Scope: 3 years of data, ~ 1000 ts

Evaluation criteria:

- punched (better than) **benchmark2 – evaluation 6 to 8**
- - punched (better than) **benchmark1 – score 8 to 10**
- **Starting code: https://github.com/aromanenko/ATSF/blob/wip/hw3_solution_example.ipynb**

Sum Up

The choice of a metric for measuring the accuracy of forecasts can seriously affect the forecast result.

Forecastability Index (Classification of Demand Forecasting Units to Forecastability Groups) helps to monitor forecast accuracy

The statement and solution of the clustering problem vary greatly depending on the purpose of applying its results.

Self Check Questions

Provide business advantages and disadvantages of WAPE and WAPPE error metrics

Give at least 3 examples of environmental conditions that could change the accuracy value of the demand forecast

Please provide at least two business problems in Retail that use the results of store clusterization?

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